
On the Limits of Biased Derivative Information for Nonconvex Stochastic Optimization

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Abstract

1 We consider the problem of finding δ -stationary points, i.e., $x \in \mathbb{R}^d$ such that
2 $\|\nabla F(x)\| \leq \delta$, for smooth, non-convex objectives, where the derivative oracles
3 are not only stochastic but also biased. In the first-order setting, we provide tight
4 lower bounds for finding an $O((\epsilon + B^2)^{1/2})$ -stationary point, where B is a bound
5 on the gradient bias, matching the upper bounds of Ajalloeian and Stich (2020).
6 We then establish bias-dependent lower bounds for algorithms that use higher-
7 order derivative information for finding $O(\epsilon + B)$ -stationary points, where B is
8 a bound on the maximum bias for all derivatives. To complement these lower
9 bounds, we develop trust-region based methods that, for certain ranges of bias,
10 provide guarantees that match the corresponding lower bounds. We further improve
11 upon the oracle complexity in high bias settings through a higher order variance
12 reduction scheme, in particular demonstrating the benefits, in some cases, of using
13 higher-order derivative information, while such improvements are known to be
14 unattainable for unbiased settings.

15 1 Introduction

16 For a smooth function $F : \mathbb{R}^d \rightarrow \mathbb{R}$ which has Lipschitz continuous derivatives, and has bounded
17 suboptimality Δ such that $F(0) - \inf_{x \in \mathbb{R}^d} F(x) \leq \Delta$, we focus on the task of finding an ϵ -stationary
18 point: that is, $x \in \mathbb{R}^d$ such that

$$\|\nabla F(x)\| \leq \epsilon$$

19 for some precision parameter $\epsilon > 0$. Finding stationary points is a task that has been explored in
20 numerous previous works (e.g., [10, 11]) and serves as a natural proxy for finding approximate local
21 optima.

22
23 When working with smooth, but potentially nonconvex functions, finding global optima has
24 been shown to be intractable. In fact, it was shown that for functions F whose p derivatives are all
25 smooth, the worst case oracle complexity of finding a point x such that $f(x) \leq f(x^*) + \epsilon$ scales at
26 least as $(1/\epsilon)^{d/p}$, where d is the dimensionality of the problem [23]. Therefore, we naturally turn to
27 finding local optima whose gradient norm is sufficiently small. Moreover, just like [23], we refer to
28 oracle complexity as the number of queries to derivative oracles, where the i^{th} order derivative oracle
29 returns an i^{th} derivative estimate of F at a query point x .

30
31 There has been a collection of work that has studied the oracle complexity of finding ϵ -
32 stationary points (i.e. $\mathbb{E}\|\nabla F(x)\| \leq \epsilon$). In [17], the authors derive an $O(\epsilon^{-4})$ oracle complexity
33 bound for using first order methods (SGD) to find an ϵ -stationary point. This first order complexity
34 bound was improved in [16] to $O(\epsilon^{-3.5})$, with the additional assumption that the stochastic gradient
35 $\nabla F(x, \xi)$ was L -smooth. One can also further improve this complexity bound to $O(\epsilon^{-3})$ if the noisy

36 gradient satisfies a mean-squared smoothness property [6]. Moreover, after incorporating access to a
 37 stochastic Hessian $\nabla^2 F(x, \xi)$, we also get a complexity bound $O(\epsilon^{-3.5})$, while also relaxing the
 38 smoothness assumption of the stochastic gradient [25]. There has also been several works which
 39 employ variance reduction (e.g. [15, 26]), some of which use methods like hessian-vector products to
 40 compute better representations of the gradient and have an even better $O(\epsilon^{-3})$ oracle complexity.

41
 42 These works are part of a broader literature that assume access to stochastic and unbiased
 43 derivative oracles, where for all derivatives $i = 1, \dots, p$, we have that

$$\mathbb{E}[\widehat{\nabla}^i F(x, \xi)] = \nabla^i F(x) \quad \text{and} \quad \mathbb{E}[\|\widehat{\nabla}^i F(x, \xi) - \nabla^i F(x)\|_{\text{op}}^2] \leq \sigma_i^2$$

44 for some set of variance parameters $\sigma_1, \dots, \sigma_p$ and where the noise ξ is drawn from some distribution
 45 P_ξ .

46 However, in many settings it becomes necessary to relax these unbiased derivative assumptions,
 47 for example in distributed [14, 4, 5], gradient-free [24], and bandit convex optimization [18], thus
 48 highlighting the need to better understand the limits of working with biased derivative information.

49 In this paper, we establish the limits of biased derivative information for both first and higher-order
 50 settings. We first complement the first-order upper bound in [3] by providing a matching lower
 51 bound. We then show how to handle biased and stochastic *high-order* derivative information, through
 52 providing corresponding higher order lower bounds and developing higher order trust-region and
 53 variance-reduction based algorithms to complement these lower bounds. We show that, unlike in the
 54 *unbiased* case [7], appealing to higher order information beyond second-order information *can* offer
 55 benefits for certain ranges of bias. In particular, we consider the following oracle model, where for
 56 all derivatives $i = 1, \dots, p$, we have that

$$\tilde{\nabla}^i F(x, \xi, b) = \nabla^i F(x) + \xi_i(x, z) + b_i(x)$$

57 where

$$\mathbb{E}_{z \sim P_z}[\xi_i(x, z)] = 0 \quad \text{and} \quad \mathbb{E}[\|\xi_i(x, z)\|_{\text{op}}^2] \leq \sigma_i^2 \quad \text{and} \quad \|b_i(x)\|_{\text{op}} \leq B_i$$

58 for some set of variance parameters $\sigma_1, \dots, \sigma_p$ and bias parameters $B_1, \dots, B_p$.

59 1.1 Our Main Contributions

60 We build on previous works in stochastic nonconvex optimization, by now considering a setting
 61 where our derivative oracles are biased as well stochastic. Below, we outline the main contributions
 62 of this paper.

63 **First-Order Lower Bound.** In [3], the authors derived the following upper bound for the oracle
 64 complexity for finding iterates $\{x_t\}$ such that $\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E}[\|\nabla F(x^{(t)})\|^2] = O(\epsilon + B_1^2)$:

$$O\left(\frac{\Delta L_1}{\epsilon + B_1^2} + \frac{\Delta L_1 \sigma_1^2}{\epsilon^2 + B_1^4}\right)$$

65 which is equivalent to finding an $O((\epsilon + B_1^2)^{\frac{1}{2}})$ stationary point, where L_1 is the Lipschitz constant
 66 of ∇F . We derive a matching lower bound that matches the provided upper bound up to constant
 67 factors to demonstrate that the upper bound is tight. This is shown in Theorem 1.

68 **Higher-Order Lower Bound.** To understand how algorithms that use derivative orders $p \geq 2$
 69 would behave in the worst case scenario, we derive the following worst case oracle complexity for
 70 finding $\epsilon + \max_i B_i$ -stationary points:

$$\Omega(1) \cdot \frac{(\sigma_1^2 - 4(\epsilon + B)^2 B_1^2) \Delta}{32(\epsilon + B)^3 \ell_0^2 \Delta_0} \cdot \min_{q' \in \{1, \dots, p\}, q \in \{2, \dots, p\}} \min \left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B) \ell_{q'}} \right)^{\frac{1}{q'}} \right\}$$

71 where $B = \max_i B_i$. This is shown in Theorem 2. To check if these lower bounds were tight, we
 72 developed algorithms to try and match these lower bounds as closely as possible.

73 **Minibatch Derivative Estimation.** We develop an algorithm where each derivative estimate D^i
 74 can be computed as an average of n_i calls to the i^{th} derivative oracle:

$$D^i(x) = \frac{1}{n_i} \sum_{j=1}^{n_i} \tilde{\nabla}^i F(x, \xi_j, b_i)$$

75 where $\tilde{\nabla}^i F$ is a biased and stochastic i^{th} derivative oracle. At each step, we solve the following
 76 subproblem:

$$x^{(t+1)} = \underset{y: \|y - x^{(t)}\| \leq \eta}{\operatorname{argmin}} \sum_{i=1}^p \frac{1}{i!} D^{(i)}[y - x^{(t)}]^i + \frac{M}{(p+1)!} \|y - x^{(t)}\|^{p+1}$$

77 for some $M \geq 8L_p$ where L_p is the Lipschitz constant of $\nabla^p F$. We found that as $p \rightarrow \infty$, the oracle
 78 complexity for finding an $O(\epsilon + \max_i B_i)$ stationary point worsens, so we run the above scheme for
 79 $p = 2$. See Theorem 3 for a more detailed description.

80 **Variance Reduction Based Derivative Estimation.** Given numerous previous works which show
 81 the advantages of using variance reduction in derivative estimation, we also utilize variance reduction
 82 based techniques with hopes of improving the bias restrictions and the oracle complexity bound
 83 for finding ϵ -stationary points. We develop an improved oracle complexity bound for finding
 84 $O(\epsilon + \max_i B_i)$ stationary points in the constant bias setting, as well as provide a bound for finding a
 85 $O((\epsilon^2 + (\max_i B_i)^2)^{\frac{1}{2}}(\epsilon + \max_i B_i))$ stationary point for the high bias setting. Furthermore, unlike
 86 the previous setting, we do see benefits for appealing to higher order derivative information for the
 87 constant bias setting. See Theorem 4 for a more detailed description.

88 1.2 Additional Related Works

89 **Biased Gradient Methods.** Here, we briefly discuss some prior work that has been done relating
 90 to biased gradient methods. In [19], the authors proposed a biased SGD algorithm and analyzed
 91 the sample complexities for convex and nonconvex objectives. In [8], the authors explore a balance
 92 between biased and unbiased estimation of the gradient to resolve the tradeoff between the cost
 93 and benefit of computing an unbiased derivative estimation. In [13, 21], the authors also present
 94 algorithms under biased gradient estimation. In [12], the authors present an analysis of biased SGD
 95 is convex and nonconvex settings, under weaker assumptions than many previous works in this area.

96 **Deterministic Oracles.** We briefly discuss additional related works that give some more broader
 97 context for our work. First, we discuss several known rates for finding ϵ -stationary points for
 98 nonconvex objectives, where the oracles are deterministic (i.e. noiseless and unbiased). First, an
 99 improvement $O(\epsilon^{-\frac{7}{4}})$ to the previously known $O(\epsilon^{-2})$ query complexity for first order methods
 100 was achieved in [9] through incorporating second order information and assuming that the Hessian
 101 is Lipschitz continuous, where the lower bound for deterministic algorithms that only rely on first
 102 and second order information is $\Omega(\epsilon^{-\frac{12}{7}})$ [11]. Furthermore, the authors highlight how this query
 103 complexity can be improved by appealing to higher order information. In particular, when using p^{th}
 104 order oracles (assuming that all p derivatives are Lipschitz continuous), we get an oracle complexity
 105 of $O(\epsilon^{(-1-\frac{1}{p})})$, thereby yielding an $O(\epsilon^{-1})$ complexity as $p \rightarrow \infty$. Moreover, in [2], the authors
 106 present a method that uses a cubic regularized Newton step to achieve an $\tilde{O}(\epsilon^{-\frac{7}{4}})$ oracle complexity
 107 for finding an ϵ -stationary point x that also satisfies $\nabla^2 f(x) \succeq -\epsilon^{\frac{1}{2}} I$.

108 1.3 Paper Organization

109 We formally go introduce our problem setup in section 2, including the function class and derivative
 110 oracle properties. In section 3, we present the lower bounds for both the first and higher order settings.
 111 In section 4, we present algorithms that use minibatch-based derivative estimation (1) and variance
 112 reduction based derivative estimation (3). We conclude the paper in section 5. We prove Theorem 3
 113 in appendix A, Theorem 4 in appendix B, and Theorems 1 and 2 in appendix C.

114
 115 **Notation.** For some $1 \leq i \leq p$, let $\nabla^i F$ refer to the i^{th} derivative of a function $F \in \mathcal{C}^p$, where $\mathcal{C}^p$ de-
 116 notes the set of p times differentiable, continuous functions. For all i , $[\nabla^i F(x)]_{j_1, \dots, j_i} = \frac{\partial^i F}{\partial x_{j_1} \dots \partial x_{j_i}}$.

For matrices A and tensors T , $\|\cdot\|_{\text{op}}$ denotes the operator norm, and unless otherwise specified $\|\cdot\|$ refers to the operator norm. For a symmetric tensor T , we let $\|T\|_{\text{op}} = \sup_{\|v\|=1} |\langle T, v, \dots, v \rangle|$. We also let $B = \max_{1 \leq i \leq p} B_i$.

2 Setup and Background

2.1 Function Class

We consider smooth, differentiable functions in the following function class:

$$\mathcal{F}_p(\Delta, L_{1:p}) = \{F : \mathbb{R}^d \rightarrow \mathbb{R} : \|\nabla^q F(x) - \nabla^q F(y)\| \leq L_q \|x - y\| \forall x, y \in \mathbb{R}^d, q \in \{1, \dots, p\}\}$$

where for all $F \in \mathcal{F}_p(\Delta, L_{1:p})$, we have that

$$F(0) - \inf_{x \in \mathbb{R}^d} F(x) \leq \Delta$$

2.2 Oracles

For such a function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, we consider a class of biased and stochastic derivative oracles for $\nabla^1 F, \dots, \nabla^p F$ defined by a distribution P_z over a measurable set $\mathcal{Z}$ and an estimator

$$O_F^p(x, z, b) := (\tilde{F}(x, z, b_0), \tilde{\nabla} F(x, z, b_1), \dots, \tilde{\nabla}^p F(x, z, b_p))$$

where $\tilde{\nabla}^q F(x, z, b_q)$ is a biased and stochastic estimate for $\nabla^q F(x)$. For all x and $q \in [p]$, we have that $\tilde{\nabla}^q F(x, z, b) = \nabla^q F(x) + \xi_q(x, z) + b_q(x)$, where $\mathbb{E}_{z \sim P_z}[\xi_q(x, z)] = 0$, $\mathbb{E} \|\xi_q(x, z)\|^2 \leq \sigma_q^2$, and $\|b_q(x)\| \leq B_q$. Given variance parameters $\sigma_{1:p}$ and bias parameters $B_{1:p}$, we define the oracle class $\mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ to be the set of all biased and stochastic p^{th} order oracles such that the conditions above hold.

3 Lower Bounds

We first consider the scenario of finding an $O(f(\epsilon) + g(B_1, \dots, B_p))$ stationary point, where f and g are positive functions of the precision parameter ϵ and the bias terms respectively. In [3], the authors present the following upper bound on the number of oracle queries for finding iterates $\{x_t\}$ where $\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla F(x^{(t)})\|^2 = O(\epsilon + B_1^2)$ (equivalently finding $x \in \{x_t\}$ such that $\|\nabla F(x)\| = O((\epsilon + B_1^2)^{\frac{1}{2}})$):

$$O\left(\frac{\Delta L_1}{\epsilon + B_1^2} + \frac{\Delta L_1 \sigma_1^2}{\epsilon^2 + B_1^4}\right)$$

In Theorem 1, we derive a matching lower bound that matches the aforementioned upper bound.

Theorem 1. *When $p = 1$, there exists $F \in \mathcal{F}_1(\Delta, L_1)$ and $(O_F^1, P_z) \in \mathcal{O}_1(F, \sigma_1, B_1)$ such that for any first-order zero-respecting algorithm (definition 1) where $\epsilon < \frac{1}{4}$ and $B_1 \leq O(1)$, the minimum number of queries to obtain a $(\epsilon + B_1^2)^{\frac{1}{2}}$ stationary point with constant probability is bounded below by*

$$\Omega\left(\frac{\Delta L_1}{\epsilon + B_1^2} + \frac{\Delta L_1 \sigma_1^2}{\epsilon^2 + B_1^4}\right)$$

Given the matching upper and lower bounds in the first order setting, one natural question was whether one could observe analogous lower and upper bounds in higher order settings. In Theorem 2, we derive a lower bound for finding an $O(\epsilon + B)$ stationary point using derivatives $1, \dots, p$ for $B \leq \frac{\sqrt{3}}{2} \sigma_1$. The function F that we will use is a rescaling of the function F_T defined on line 1. In Lemma 10, we prove that for all $x, y \in \mathbb{R}^d$, there exists a constant $C \geq 0$ such that $\|\nabla F_T(x)\| \leq C\sqrt{T}$. Since we scale this function as the following $F_T^*(x) = \alpha F_T(\beta x)$ for constants α, β , we have that $\|\nabla F_T(x)\| \leq C_1\sqrt{T}$ for a different C_1 . For $B \geq C_1\sqrt{T} - \epsilon$, we need zero oracle queries to reach such the given stationarity condition, as for any $x \in \mathbb{R}^d$, we have that $\|\nabla F(x)\| \leq C_1\sqrt{T}$, so the lower bound is vacuous for this scenario.

Theorem 2. For all $p \geq 2$, Δ , $L_{1:p}$, $\sigma_{1:p} > 0$, $\epsilon < \sqrt{\sigma_1}$, and $B \leq \frac{\sqrt{3}}{2}\sigma_1$, there exists $F \in \mathcal{F}_p(\Delta, L_{1:p})$ and $(O_F^p, P_z) \in \mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ such that for any p^{th} order zero-respecting algorithm (definition 1), the number of queries to obtain a point an $\epsilon + \max_i B_i$ stationary point with constant probability is bounded below by

$$\Omega(1) \cdot \frac{(\sigma_1^2 - 4(\epsilon + B)^2 B_1^2) \Delta}{32(\epsilon + B)^3 \ell_0^2 \Delta_0} \cdot \min_{q' \in \{1, \dots, p\}, q \in \{2, \dots, p\}} \min \left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B) \ell_{q'}} \right)^{\frac{1}{q'}} \right\}$$

Proof Sketch. Here, we provide a brief proof sketch, deferring the full proofs to appendix C. We outline the proof of Theorem 2, as the proof of Theorem 1 will follow a similar argument to this one. We first introduce the following “hard” function [10] where for a fixed $T \geq 0$ and $x \in \mathbb{R}^T$:

$$F_T(x) = -\Psi(1)\Phi(1) + \sum_{i=2}^T [\Psi(-x_{i-1})\Phi(-x_i) - \Psi(x_{i-1})\Phi(x_i)]$$

where

$$\Psi(x) = \begin{cases} 0, & \text{if } x \leq \frac{1}{2}, \\ \exp(1 - \frac{1}{(2x-1)^2}), & \text{if } x > \frac{1}{2}, \end{cases}, \Phi(x) = \sqrt{e} \int_{-\infty}^x e^{-\frac{1}{2}t^2} dt, \quad (1)$$

and we let ℓ_p represent the Lipschitz constant of the p^{th} derivative of F_T . We then show how to construct the following series of derivative estimators for this function (for all derivatives $i \in \{1, \dots, p\}$):

$$[\tilde{\nabla}^q F_T(x, z, b)]_i = (1 + \mathbf{1}\{i > \text{prog}_{\frac{1}{4}}(x)\}(\frac{z}{\rho} - 1)) \cdot (\nabla_i^q F_T(x) + b_i^q(x))$$

which crucially account for the fact that the oracles are not only stochastic *but also biased*. If T is a k -dimensional tensor, then T_i is a $k-1$ -dimensional subtensor where $[T_i]_{j_1, \dots, j_{k-1}} = T_{i, j_1, \dots, j_k}$ and $\text{prog}_{\alpha}(x) = \max\{i \geq 0, |x_i| > \alpha\}$, representing the highest index of $x \in \mathbb{R}^d$ that is at least α away from zero. We carefully devised the above construction such that this collection of derivative estimators formed a probability- ρ zero-chain (for some $0 \leq \rho \leq 1$) where the following properties hold:

$$\begin{aligned} \Pr(\exists x | \text{prog}(\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)) = \text{prog}_{\frac{1}{4}}(x) + 1) &\leq \rho \\ \Pr(\exists x | \text{prog}(\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)) = \text{prog}_{\frac{1}{4}}(x) + i) &= 0 \end{aligned}$$

for all $i > 1$. Through this zero-chain construction (as is well studied in [1, 10, 11, 6]), we enforce that every oracle query can reveal information about at most one new coordinate, thereby requiring any algorithm to make sequential progress, which yields a lower bound on the number of queries to make sufficient progress (see appendix C for a more formal explanation). Given that we have shown that $\{\tilde{\nabla}^q F_T(x, z, b)\}$ form a probability- ρ zero-chain, we used a scaled version of F_T , parametrized by two constants α and β :

$$F_T^*(x) = \alpha F_T(\beta x)$$

where we solved for α and β to enforce the required suboptimality, noise, bias, and higher order Lipschitz conditions. $\square$

4 Upper Bounds

4.1 Minibatch-Based Derivative Estimation

Given these lower bounds, we develop various algorithms to how the upper bounds compared. In Algorithm 1, we minimize a regularized second order model of our objective F at each iteration. In Theorem 3, we prove an upper bound for the oracle complexity of Algorithm 1 for reaching an $O(\epsilon + B)$ stationary point.

Algorithm 1 Biased and Stochastic Cubic-Regularized Trust Region

Require: Biased and stochastic oracle $(O_F^p, P_z) \in \mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ for $F \in \mathcal{F}_p(\Delta, L_{0:p})$, Precision parameter ϵ , Initial parameter $x^{(0)}$

1: Find constants $\{C_i\}_{i=1}^p$ such that (for all n and all $t \geq 1$):

$$\mathbb{E} \|D_t^{(i)} - \nabla F^{(i)}(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}} \leq 2^{1/p} \cdot \left(\left(\frac{C_i \cdot \sigma_i^2}{n} \right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{p}} \right)$$

2: $M \leftarrow 8L_p, \eta \leftarrow \min\{(\epsilon + \max_i B_i)^{\frac{1}{p}}, 1 - \epsilon\}$

3: $A \leftarrow \frac{16(p+1)!}{M} \max\{1, \frac{(p!)^{\frac{p+1}{p}}}{4M^{\frac{1}{p}}} + 2(\frac{2p!}{M})^{\frac{1}{p}}, \frac{(p!)^{\frac{p+1}{p}}}{4M^{\frac{1}{p}}} + 2(\frac{2p \cdot p!}{M})^{\frac{1}{p}}\}$

4: $T \leftarrow \lceil \frac{8A\Delta}{\eta^{p+1}} \rceil$

5: Pick n_1 such that

$$\max\left\{ \frac{C_1 \cdot \sigma_1^2}{\left(\frac{\eta^{p+1}}{8A} - B_1^{\frac{p+1}{2p}} \right)^{\frac{2p}{p+1}}}, 1 \right\} \leq n_1 \leq \frac{(\epsilon + \max_i B_i)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (1 + \max_i B_i)^{\frac{p+1}{p}}}$$

6: For all $2 \leq i \leq p$, pick n_i such that

$$\max\left\{ \frac{C_i \sigma_i^2}{\left(\frac{\eta^{\frac{p^2-1}{p}}}{8Ap} - B_i^{\frac{p+1}{2p}} \right)^{\frac{2p}{p+1}}}, 1 \right\} \leq n_i \leq \frac{(\epsilon + \max_i B_i)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (1 + \max_i B_i)^{\frac{p+1}{p}}}$$

7: **for** $t = 0$ to $T - 1$ **do**

8: **for** $i = 1$ to p **do**

9: Query the i^{th} order oracle n_i times at $x^{(t)}$ and compute

$$D^i(x^{(t)}) = \frac{1}{n_i} \sum_{j=1}^{n_i} \tilde{\nabla} F(x^{(t)}, z^{(t,j)}, b_i), \quad z^{(t,j)} \sim P_z$$

10: **end for**

11: Set the next point $x^{(t+1)}$ as

$$x^{(t+1)} = \underset{y: \|y - x^{(t)}\| \leq \eta}{\operatorname{argmin}} \sum_{i=1}^p \frac{1}{i!} D^{(i)}[y - x^{(t)}]^i + \frac{M}{(p+1)!} \|y - x^{(t)}\|^{p+1}$$

12: **end for**

13: **return** $\hat{x}$ chosen uniformly at random from $\{x^{(t)}\}_{t=1}^T$

183 **Theorem 3.** For any function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, where $p \geq 2, \epsilon > 0$, with biased and stochastic
184 p^{th} -order oracles in $\mathcal{O}(F, \sigma_{1:p}, B_{1:p})$ where $B \geq \Omega(\epsilon^{\frac{3p}{3p+1}})$, with probability at least $\frac{5}{8}$, Algorithm 1
185 returns a point $\hat{x}$ such that $\|\nabla F(\hat{x})\| \leq O(\epsilon + B)$ and performs at most

$$O\left(\frac{\Delta(\max_i \sigma_i)^2}{\epsilon^3 (B+1)^{\frac{p+1}{p}}} + \frac{(\epsilon + B)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (B+1)^{\frac{p+1}{p}}} \right)$$

186 queries to the stochastic and biased derivative oracles.

187 Here, the fact that we require $B \geq \Omega(\epsilon^{\frac{3p}{3p+1}})$ is due to requirements of the algorithm in terms of batch
188 size and that the existence of constants $\{C_i\}$ is guaranteed by Lemma 1. Note in the unbiased case
189 (i.e $B = 0$), we recover the $O(\epsilon^{-3})$ guarantee known from [7]. Moreover, as $p \rightarrow \infty$, the oracle
190 complexity worsens, thereby implying that using derivative information beyond the second order
191 does not help in this scenario. Therefore, in the minibatch derivative estimation setting, setting $p = 2$
192 yields optimal oracle complexity as compared to any $p > 2$. Interestingly, however, when using
193 variance reduction, we do realize benefits to appealing to higher order derivative information for
194 certain ranges of bias.

195 *Proof Sketch.* Here, we provide a brief proof sketch of Theorem 3, deferring the full proof to appendix
 196 A. In Lemma 1, we prove that there do exist constants C_i that satisfy the condition on line 1 of
 197 Algorithm 1. We then derive a lower bound for $F(x) - F(y)$ in Lemma 3 for all $M \geq 8L_p$ and
 198 $0 \leq \eta < 1$, where $x \in \mathbb{R}^d$ and $y \in \operatorname{argmin}_{z: \|z-x\| \leq \eta} m_x(z)$, where m_x represents the regularized
 199 p^{th} order model of F around x using biased and stochastic derivatives:

$$m_x(y) = F(x) + \sum_{i=1}^p \frac{1}{i!} D^{(i)}[y-x]^i + \frac{M}{(p+1)!} \|y-x\|^{p+1}$$

200 We then extend this Lemma to the case where $D^{(i)}$ are random variables in Lemma 5, and then
 201 (through using an intermediary Lemma 6)) prove an upper bound of $\frac{3}{8}$ on the probability of reaching
 202 a point $\hat{x}$ such that $\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}(\epsilon + B))$, which completes the proof. $\square$

203 Below, we also provide the comparison to the lower bound in Theorem 2 in terms of the oracle
 204 complexity.

Bias Regime	Lower Bound (Theorem 2)	Upper Bound (Theorem 3)
$B \leq O(\epsilon)$	$\Omega(\epsilon^{-3})$	$O(\epsilon^{-3})$
$B = \Theta(1)$	$\Omega(1)$	$O(\epsilon^{-3})$
$B = \Theta(\epsilon^{-q}), q > 0$	Trivial	$O(\epsilon^{\frac{q(p+1)-3p}{p}} + \epsilon^{-3})$

205 This demonstrates that the lower bound in Theorem 2 is tight for $B \leq O(\epsilon)$. We work on improving
 206 these bounds through a variance reduction scheme (Algorithm 3), succeed in improving upon the
 207 upper bound for the $B = \Theta(1)$ case to $O(\epsilon^{-2})$, and leave further improvements for future work. We
 208 note that if one knew apriori that $B = \Theta(\epsilon^{-q})$ for $q > 0$, then one can directly return the starting
 209 iterate $x^{(0)}$, so in theory, one could match the lower bound for this setting as well. We do not assume
 210 such knowledge in our algorithm.

211 4.2 Variance Reduction

212 Given the numerous prior works that have demonstrated the advantages of using variance reduction
 213 for derivative estimation, we investigate the potential advantages of using variance reduction in a
 214 biased setting as well. Many previous works have primarily relied on recursive variance reduction
 215 (e.g. [15]) to compute cheap estimators of the gradient $\nabla F(x^{(t)})$. In our implementation of recursive
 216 variance reduction, we build on that of [7] by estimating $\nabla^i F(x^{(t)}) - \nabla^i F(x^{(t+1)})$ by averaging
 217 $\nabla^{i+1} F$ -vector products for all $i \in [p]$, instead of just doing this with the gradient. To derive this
 218 estimator, we first note that for all i , it holds that (by the Fundamental Theorem of Calculus) for
 219 all x, x' : $\nabla^i F(x) - \nabla^i F(x') = \int_0^1 \nabla^{i+1} F(xt + x'(1-t))(x - x')dt$. Now, to approximate this
 220 integral, we construct the following estimator for $\nabla^i F$, where K is chosen to be proportional to
 221 $\|x - x'\|^2$:

$$\tilde{\nabla}^i F = \frac{1}{K} \sum_{k=0}^{K-1} \tilde{\nabla}^{i+1} F(x \cdot (1 - \frac{k}{K}) + x' \cdot \frac{k}{K}, z^{(i)}, b_i)(x - x')$$

222 We reset the derivative estimators according to a defined probability metric b and dynamically set the
 223 batch size proportional to the difference between the current iterate and the previous iterate squared
 224 and incorporate this recursive variance reduction approach for all p derivatives. In Theorem 4, we
 225 analyze the oracle complexity of a variance-reduction based algorithm (Algorithm 2) for finding an
 226 $O(\epsilon + B)$ stationary point and a $O((\epsilon^2 + B^2)^{\frac{1}{2}}(\epsilon + B))$ stationary point.

227 **Theorem 4.** For any function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, with biased and stochastic p^{th} order oracles in
 228 $\mathcal{O}(F, \sigma_{1:p}, B_{1:p})$, with probability at least $\frac{5}{8}$, Algorithm 3 returns a point $\hat{x}$ such that:

- 229 • If $B = \Theta(1)$, then $\|\nabla F(\hat{x})\| \leq O(\epsilon + B)$ with at most

$$O\left(\frac{\Delta(\max_i \sigma_i)^2(\epsilon + B)^{\frac{1}{p}} + (\max_i \sigma_i)^2}{\epsilon^2} + \frac{\Delta(\epsilon + B)^{\frac{1}{p}} + 1}{\epsilon}\right)$$

230 queries to the stochastic and biased derivative oracles.

Algorithm 2 Higher-Order Recursive Variance Reduction (HO-RVR)

Require: Precision parameter ϵ , probability b , current iterate x , previous iterate x_{prev} , derivative order i , derivative estimate with respect to x_{prev} , D_{prev}^i , Biased and stochastic oracle $(O_F^p, P_z) \in \mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ for $F \in \mathcal{F}_p(\Delta, L_{1:p})$

1: Set

$$K = \left\lceil \frac{5(\sigma_{i+1}^2 + L_{i+1}\epsilon)}{b\epsilon^2} \cdot \|x - x_{\text{prev}}\|^2 \right\rceil$$

2: Set $n = \left\lceil \frac{5\sigma_i^2}{\epsilon^2} \right\rceil$

3: Sample $C \sim \text{Bernoulli}(b)$.

4: **if** C is 1 **or** D_{prev}^i is None **then**

5: Query the i^{th} order oracle n times at x and set

$$D^{(i)} = \frac{1}{n} \sum_{j=1}^n \tilde{\nabla}^i F(x, z^{(j)}, b_i), \quad z^{(j)} \sim P_z$$

6: **else**

7: For $k \in \{0, \dots, K\}$, set

$$x^{(k)} = \frac{k}{K}x + (1 - \frac{k}{K})x_{\text{prev}}$$

8: Query the i^{th} order oracle at the points $\{x^{(k)}\}_{k=0}^{K-1}$ and set

$$D^{(i)} = D_{\text{prev}}^{(i)} + \sum_{k=1}^K \tilde{\nabla}^{i+1} F(x^{(k-1)}, z^{(k)}, b_{i+1}), \quad z^{(k)} \sim P_z$$

9: **end if**

10: **return** $D^{(i)}$

231 • If $B > \Omega(1)$, then $\|\nabla F(\hat{x})\| \leq O((\epsilon^2 + B^2)^{\frac{1}{2}}(\epsilon + B))$ with at most

$$O\left(\frac{(\max_i \sigma_i)^2}{\epsilon^2(\epsilon + B)^{\frac{p+1}{p}}} + \frac{1}{\epsilon(\epsilon + B)^{\frac{p+1}{p}}}\right)$$

232 queries to the stochastic and biased derivative oracles.

233 *Proof Sketch.* Here, we provide a brief proof sketch, deferring the full proof to appendix B. In Lemma
234 7, we prove that

$$\mathbb{E}\|D^{(i)}(x^{(t)}) - \nabla^i F(x^{(t)})\|^2 \leq 4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{\frac{2}{p}} + 18B^2$$

235 thereby establishing a bound on the difference in the derivative estimate versus the true derivative for
236 all derivative orders. We then derive an upper bound for $\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p)$ in terms of B, ϵ ,
237 and the hyperparameters of Algorithm 3 in Lemma 8. Plugging the parameters in gives an upper
238 bound of $\frac{3}{8}$ for $\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p)$, which finishes the proof. $\square$

239 Notably, from Theorem 4, one can observe the following important conclusions:

- 240 • Unlike the purely minibatch-based approach for derivative estimation (utilized in Algorithm
241 1), appealing to higher order information does provide advantages in the event where
242 $B = \Theta(1)$.
- 243 • In the $B = \Theta(1)$ setting, we have a significantly improved oracle complexity of $O(\epsilon^{-2})$
244 compared to the $O(\epsilon^{-3})$ complexity from Algorithm 1.

Algorithm 3 Higher-Order Recursive Variance Reduction Derivative Estimation (HO-RVR-D)

Input: Precision parameter ϵ , Biased and stochastic oracle $(O_F^p, P_z) \in \mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ for $F \in \mathcal{F}_p(\Delta, L_{1:p})$, derivative order p .

- 1: Pick b such that $0 < b \leq 1$, let $B = \max_i B_i$
- 2: Let $X = 4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{\frac{2}{p}} + 18B^{\frac{2}{p}}$
- 3: Set $\eta = \min\{(\epsilon + B)^{\frac{1}{p}}, 1 - \epsilon\}$
- 4: Let $A = \max(16(p+1)!, 2(p+1)! \cdot [(p!)^{\frac{p+1}{p}} + 8 \cdot (2p!)^{\frac{1}{p}}], 2(p+1)! \cdot [(p!)^{\frac{p+1}{p}} + 4(2p \cdot p!)^{\frac{1}{p}}])$
- 5: Set $M = \max\{(\frac{8AX^{\frac{p+1}{2p}}}{\eta^{p+1}})^{\frac{p}{p+1}}, (\frac{8Ap \cdot X^{\frac{p+1}{2p}}}{\eta^{\frac{p^2-1}{p}}})^{\frac{p}{p+1}}, (\epsilon + B)^{\frac{-p-2}{p}}, 8L_p\}$
- 6: Set $T = \lceil \frac{8A\Delta}{M\eta^{p+1}} \rceil$
- 7: Set $x^{(0)} = x^{(1)} = 0$, $D^{(i)} = \text{None}$ for $i \in \{1, \dots, p\}$
- 8: **for** $t = 1$ to T **do**
- 9: $D_t^{(i)} = \text{HO-RVR}(\epsilon, b, x^{(t)}, x^{(t-1)}, D_{t-1}^{(i)})$
- 10: Set the next point $x^{(t+1)}$ as

$$x^{(t+1)} = \underset{y: \|y - x^{(t)}\| \leq \eta}{\operatorname{argmin}} \sum_{i=1}^p \frac{1}{i!} D_t^{(i)} [y - x^{(t)}]^i + \frac{M}{(p+1)!} \|y - x^{(t)}\|^{p+1}$$

11: **end for**

12: **return** $\hat{x}$ chosen uniformly at random from $\{x^{(t)}\}_{t=2}^{T+1}$

- Consider the scenario where $B > \Omega(1)$. Unlike the previous scenario, appealing to higher order derivatives does not yield a better oracle complexity. Our intuition is to why this is the case, is if $B > \Omega(1)$, the stationary guarantee is quite weak, implying that there may be little to no difference between appealing to higher order information and not doing so in terms of oracle complexity. Moreover, for the $p = 2$ case, we have an improved oracle complexity from the variance reduction based scheme in Algorithm 2 compared to Algorithm 1 for all settings of $\{B_i\}$ such that $B > \Omega(1)$.

5 Conclusion

This paper extends the settings of deterministic derivative oracles and stochastic but unbiased oracles to consider derivative oracles that are both stochastic and biased. We provide a matching first order lower bound to complement the first order upper bound that is provided in this stochastic and biased scenario in [3]. We further extend this lower bound for algorithms that use second order derivative information or higher for finding $O(\epsilon + B)$ stationary points. Then, to complement these lower bounds, we developed trust region based methods, that under certain bias regimes, matches the corresponding lower bound up to constant factors. We then improved upon these algorithms by incorporating a higher order variance reduction scheme, which improves the oracle complexity for certain ranges of bias, and in some cases, reveals advantages of appealing to higher order derivative information.

With regards to opportunities for future work, our upper bound in Theorem 3 only matched the corresponding higher order lower bound in Theorem 2 for $O(\epsilon)$ bias, which leaves open the possibility of a stronger upper bound. Moreover, it would be interesting to consider additional cases where appealing to higher order derivative information would be beneficial for algorithms relying on biased and stochastic oracle access.

LLM Usage. LLMs were used as an assistive tool for checking the algebraic steps of our proofs. All algorithms, theorem statements, and final proofs were developed and verified by the authors.

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338 A Appendix A

339 **Lemma 1.** Given $A_i \in \mathbb{R}^{d_1 \times \dots \times d_m}$, where $d_1 = \dots = d_m = d$, and $\mathbb{E}[A_i] = B$ and $\mathbb{E}[\|A_i -$
 340 $B\|^2] \leq \sigma^2$, we have that

$$\mathbb{E}[\|\frac{1}{n} \sum_{i=1}^n A_i - B\|_{\text{op}}^2] \leq \frac{C \cdot \sigma^2}{n}$$

341 for some d -dependent, n -independent constant $C \geq 0$.

342 *Proof.* Let $X_i = A_i - B$ and observe that

$$\begin{aligned} \mathbb{E}[\|\sum_{i=1}^n X_i\|_{\text{op}}^2] &\leq \mathbb{E}_{X, X'}[\|\sum_{i=1}^n X_i - X'_i\|_{\text{op}}^2] \\ &= \mathbb{E}_{X, X', \epsilon}[\|\sum_{i=1}^n \epsilon_i (X_i - X'_i)\|_{\text{op}}^2] \\ &\leq 4\mathbb{E}_{X, \epsilon}[\|\sum_{i=1}^n \epsilon_i X_i\|_{\text{op}}^2] \end{aligned}$$

343 where $(X'_i)_{i=1}^n$ is a sequence of independent copies of $(X_i)_{i=1}^n$ and $(\epsilon_i)_{i=1}^n$ is a sequence of
 344 Rademacher random variables. Now, take S such that $S \subset \{1, \dots, m\}$, where $|S| = \lfloor m/2 \rfloor$.
 345 We define

$$Z_i \in \mathbb{R}^{(\prod_{k \in S} d_k) \times (\prod_{k \in S^c} d_k)}$$

346 to be a flattened version of X_i . Let $D = \min\{\prod_{k \in S} d_k, \prod_{k \in S^c} d_k\}$, so in this case, $D = d^{\lfloor m/2 \rfloor}$.
 347 We now prove that for any p , there exists d -dependent constants C_1, C_2, C_3 such that

$$\|X_i\|_{\text{op}} \leq C_2 \cdot \|Z_i\|_{S_{2p}} \leq C_2 C_1 C_3^2 \cdot D^{\frac{1}{2p}} \cdot \|X_i\|_{\text{op}}$$

348 We note that

$$\begin{aligned} \|X_i\|_{\text{op}} &= \sup_{\|u^{(1)}\|=1 \dots \|u^{(m)}\|=1} \langle X_i, u^{(1)} \otimes \dots \otimes u^{(m)} \rangle \\ &= \sup_{\|a_1\|=\|b_1\|=1} \langle Z_i, a_1 b_1^T \rangle \\ &\leq \sup_{\|a\|=\|b\|=1} \langle Z_i, a b^T \rangle \\ &= \|Z_i\|_{\text{op}} \leq C_2 \cdot \|Z_i\|_2 \leq C_2 \cdot \sigma_{\max}(Z_i) \end{aligned}$$

349 for some constant C_2 , since due to norm equivalence in finite dimensional spaces, there exists
 350 constants C_1, C_2 such that $C_1 \cdot \|Z_i\|_2 \leq \|Z_i\|_{\text{op}} \leq C_2 \cdot \|Z_i\|_2$. Also,

$$a_1 = \bigotimes_{k \in S} u^{(k)}, b_1 = \bigotimes_{k \notin S} u^{(k)}$$

351 Now

$$\begin{aligned} &C_2 \cdot \sigma_{\max}(Z_i) \\ &\leq C_2 \left(\sum_{j=1}^D \sigma_j^{2p}(Z_i) \right)^{\frac{1}{2p}} \\ &= C_2 \cdot \|Z_i\|_{S_{2p}} \\ &\leq C_2 \cdot (D \cdot \sigma_{\max}^{2p}(Z_i))^{\frac{1}{2p}} \\ &= C_2 \cdot D^{\frac{1}{2p}} \cdot \sigma_{\max}(Z_i) \\ &\leq C_2 C_1 \cdot D^{\frac{1}{2p}} \cdot \|Z_i\|_{\text{op}} \end{aligned}$$

352 Since

$$\|Z_i\|_{\text{op}} = \sup_{\|a\|=\|b\|=1} a^T Z_i b$$

353 expand a and b in their orthonormal bases as follows:

$$a = \sum_{\alpha=1}^{d^{\lfloor m/2 \rfloor}} a_{\alpha} e_{\alpha}, b = \sum_{\beta=1}^{d^{\lceil m/2 \rceil}} b_{\beta} f_{\beta}$$

354 which implies that

$$\begin{aligned} & |a^T Z_i b| \\ &= \left| \sum_{\alpha, \beta} a_{\alpha} b_{\beta} \langle X_i, e_{\alpha} \otimes f_{\beta} \rangle \right| \\ &\leq \sum_{\alpha, \beta} |a_{\alpha}| \cdot |b_{\beta}| \cdot |\langle X_i, e_{\alpha} \otimes f_{\beta} \rangle| \\ &\leq \sum_{\alpha, \beta} |a_{\alpha}| \cdot |b_{\beta}| \cdot \|X_i\|_{\text{op}} \\ &\leq \|X_i\|_{\text{op}} \cdot \|a\|_1 \cdot \|b\|_1 \\ &\leq C_3^2 \cdot \|X_i\|_{\text{op}} \end{aligned}$$

355 due to Cauchy-Schwarz inequality and since $\|x\|_1 \leq C_3 \cdot \|x\|$ for some constant C_3 for all x . This
356 implies that

$$D^{\frac{1}{2p}} \cdot \|Z_i\|_{\text{op}} \leq D^{\frac{1}{2p}} \cdot C_3^2 \cdot \|X_i\|_{\text{op}}$$

357 which proves the inequality. Now, we have that

$$\mathbb{E}_{\epsilon} \left[\left\| \sum_{i=1}^n \epsilon_i X_i \right\|_{\text{op}}^2 \right] \leq C_2^2 \cdot \mathbb{E}_{\epsilon} \left[\left\| \sum_{i=1}^n \epsilon_i Z_i \right\|_{S_{2p}}^{2p} \right]$$

358 By Matrix-Khintchine inequality [22], we have that

$$\begin{aligned} & (\mathbb{E}_{\epsilon} \left[\sum_{i=1}^n \|\epsilon_i Z_i\|_{S_{2p}}^{2p} \right])^{1/p} \\ &\leq (2p-1) \cdot \left\| \left(\sum_{i=1}^n Z_i^2 \right)^{1/2} \right\|_{S_{2p}}^2 \\ &= (2p-1) \cdot \left\| \sum_{i=1}^n Z_i^2 \right\|_{S_{2p}} \\ &\leq (2p-1) \cdot \sum_{i=1}^n \|Z_i\|_{S_{2p}}^2 \\ &\leq (2p-1) \cdot D^{1/p} C_1^2 \cdot \sum_{i=1}^n \|Z_i\|_{\text{op}}^2 \\ &\leq (2p-1) \cdot D^{1/p} C_1^2 C_3^4 \cdot \sum_{i=1}^n \|X_i\|_{\text{op}}^2 \end{aligned}$$

359 Taking $p = 1$, we have that

$$\mathbb{E}_{\epsilon} \left[\sum_{i=1}^n \|\epsilon_i Z_i\|_{S_2}^2 \right] \leq D C_1^2 C_3^4 \cdot \sum_{i=1}^n \|X_i\|_{\text{op}}^2$$

360 and when taking expectation with respect to X , we have that

$$\begin{aligned}
& \mathbb{E}[\|\sum_{i=1}^n \epsilon_i X_i\|_{\text{op}}^2] \\
& \leq C_2^2 \cdot DC_1^2 C_3^4 \cdot \sum_{i=1}^n \mathbb{E}\|X_i\|_{\text{op}}^2 \\
& \leq DC_2^2 C_1^2 C_3^4 \sigma^2 \\
& \leq d^{m/2} C_2^2 C_1^2 C_3^4 \sigma^2
\end{aligned}$$

361 Putting everything together and normalizing gives a final bound of

$$\frac{4d^{m/2} C_2^2 C_1^2 C_3^4 \sigma^2}{n}$$

362 proving the claim. $\square$

363 **Lemma 2.** For all integers $p \geq 1$, for all $i \in \{1, \dots, p\}$ and all $t \geq 1$, there exists a d -dependent,
364 n_i -independent constant $C \geq 0$ such that

$$\mathbb{E}\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}} \leq 2^{1/p} \cdot \left(\left(\frac{C \cdot \sigma_i^2}{n_i} \right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{p}} \right)$$

365 *Proof.* First, we can say that

$$\begin{aligned}
& \mathbb{E}[\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}}] \\
& = \mathbb{E}[\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)}) + b_i(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}}] \\
& \leq 2^{1/p} \cdot (\mathbb{E}[\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}}] + \mathbb{E}[\|b_i(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}}]) \\
& \leq 2^{1/p} \cdot (\mathbb{E}[\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)})\|_{\text{op}}^{\frac{p+1}{p}}] + B_i^{\frac{p+1}{p}})
\end{aligned}$$

366 Notice that for any $r \in [1, 2]$, we can have that

$$\begin{aligned}
& \mathbb{E}[\|D_t^{(i)} - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)})\|_{\text{op}}^r] \\
& = \mathbb{E}[\|\frac{1}{n_i} \sum_{j=1}^{n_i} \tilde{\nabla}^i F(x^{(t)}, z^{(t,j)}) - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)})\|_{\text{op}}^r] \\
& \leq (\mathbb{E}[\|\frac{1}{n_i} \sum_{j=1}^{n_i} \tilde{\nabla}^i F(x^{(t)}, z^{(t,j)}) - \nabla F^{(i)}(x^{(t)}) - b_i(x^{(t)})\|_{\text{op}}^2])^{r/2} \\
& \leq \left(\frac{C \cdot \sigma_i^2}{n_i} \right)^{r/2}
\end{aligned}$$

367 where we have used Lyapunov's inequality and the result from lemma 1. Thus, we have a final bound
368 of

$$2^{1/p} \cdot \left(\left(\frac{C \cdot \sigma_i^2}{n_i} \right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{p}} \right)$$

369 which finishes the proof. $\square$

370 **Lemma 3.** Given a function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, let

$$m_x(y) = F(x) + \langle D^{(1)}, y - x \rangle + \sum_{i=2}^p \frac{1}{i!} D^{(i)}(x) [y - x]^i + \frac{M}{(p+1)!} \|y - x\|^{p+1}$$

371 and let $y \in \operatorname{argmin}_{z: \|z-x\| \leq \eta} m_x(z)$ for $0 \leq \eta < 1$. Then, for all $M \geq 8L_p$, we have that

$$\begin{aligned}
& F(x) - F(y) > \frac{M}{8(p+1)!} \|y - x\|^{p+1} - \left(\frac{2p!}{M} \right)^{\frac{1}{p}} \cdot \|\nabla F(x) - D^{(1)}(x)\|_{\text{op}}^{\frac{p+1}{p}} \\
& - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \cdot \|\nabla^i F(x) - D^i(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \eta^{\frac{p+1}{p}}
\end{aligned}$$

372 *Proof.* We have that $F(y) - F(x)$

$$\begin{aligned}
&\leq F(x) + \langle \nabla F(x), y - x \rangle + \sum_{i=2}^p \frac{1}{i!} \nabla^i F(x) [y - x]^i + \frac{L_p}{(p+1)!} \|y - x\|^{p+1} - F(x) \\
&\leq m_x(y) + \langle \nabla F(x) - D^{(1)}(x), y - x \rangle + \sum_{i=2}^p \frac{1}{i!} (\nabla^i F(x) - D^{(i)}(x)) [y - x]^i + \frac{L_p - M}{(p+1)!} \|y - x\|^{p+1} - m_x(x) \\
&\leq \langle \nabla F(x) - g, y - x \rangle + \sum_{i=2}^p \frac{1}{i!} (\nabla^i F(x) - D^{(i)}(x)) [y - x]^i + \frac{L_p - M}{(p+1)!} \|y - x\|^{p+1} \\
&\leq -\frac{7M}{8(p+1)!} \|y - x\|^{p+1} + \|\nabla F(x) - g\| \cdot \|y - x\| \\
&\quad + \sum_{i=2}^p \frac{1}{i!} \|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}} \cdot \|y - x\|
\end{aligned}$$

373 since $\|y - x\| \leq \eta$ and since $\eta < 1$, we have that $\|y - x\|^{i-1} \leq \|y - x\|$ for $i \geq 2$. By Young's
374 inequality, we have that

$$\begin{aligned}
&\|\nabla F(x) - D^{(1)}(x)\| \cdot \|y - x\| \leq \left(\left(\frac{2p!}{M} \right)^{\frac{1}{p}} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \cdot \frac{p}{p+1} \right) + \left(\frac{\|y - x\|^{p+1}}{(p+1)} \cdot \frac{M}{2p!} \right) \\
&= \left(\frac{2p!}{M} \right)^{\frac{1}{p}} \left(\frac{p \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}}}{p+1} \right) + \frac{M \|y - x\|^{p+1}}{2(p+1)!}
\end{aligned}$$

375 and

$$\begin{aligned}
&\|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}} \cdot \|y - x\| \\
&\leq \left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \frac{p \cdot \|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}}^{\frac{p+1}{p}}}{p+1} + \frac{M \|y - x\|^{p+1}}{(p+1) \cdot (2p \cdot p!)} \\
&= \left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \frac{p \cdot \|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}}^{\frac{p+1}{p}}}{p+1} + \frac{M \|y - x\|^{p+1}}{2p \cdot (p+1)!}
\end{aligned}$$

376 which implies that

$$\begin{aligned}
&-\frac{7M}{8(p+1)!} \|y - x\|^{p+1} + \|\nabla F(x) - D^{(1)}(x)\| \cdot \|y - x\| \\
&+ \sum_{i=2}^p \frac{1}{i!} \|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}} \cdot \|y - x\| \\
&\leq -\frac{7M}{8(p+1)!} \|y - x\|^{p+1} + \left(\frac{2p!}{M} \right)^{\frac{1}{p}} \cdot \frac{p}{p+1} \cdot \|\nabla F(x) - g\|^{\frac{p+1}{p}} + \frac{M \|y - x\|^{p+1}}{2(p+1)!} \\
&+ \sum_{i=2}^p \frac{1}{i!} \cdot \left[\left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \frac{p \cdot \|\nabla^i F(x) [y - x, :, \dots, :] - D^{(i)}(x) [y - x, :, \dots, :]\|_{\text{op}}^{\frac{p+1}{p}}}{p+1} + \frac{M \|y - x\|^{p+1}}{2p \cdot (p+1)!} \right] \\
&\leq -\frac{3M}{8(p+1)!} \|y - x\|^{p+1} + \left(\frac{2p!}{M} \right)^{\frac{1}{p}} \cdot \frac{p}{p+1} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \\
&+ \sum_{i=2}^p \frac{1}{i!} \left[\left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \frac{p \cdot \|\nabla^i F(x) - D^{(i)}(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \|y - x\|^{\frac{p+1}{p}}}{p+1} + \frac{M \cdot \|y - x\|^{p+1}}{2p \cdot (p+1)!} \right] \\
&< -\frac{3M}{8(p+1)!} \|y - x\|^{p+1} + \left(\frac{2p!}{M} \right)^{\frac{1}{p}} \cdot \frac{p}{p+1} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \\
&+ \sum_{i=2}^p \frac{1}{2} \left[\left(\frac{2p \cdot p!}{M} \right)^{\frac{1}{p}} \frac{p \cdot \|\nabla^i F(x) - D^{(i)}(x)\|^{\frac{p+1}{p}} \cdot \|y - x\|^{\frac{p+1}{p}}}{p+1} + \frac{M \cdot \|y - x\|^{p+1}}{2p \cdot (p+1)!} \right]
\end{aligned}$$

377 We can further bound this expression by

$$\begin{aligned}
&< -\frac{3M}{8(p+1)!} \|y-x\|^{p+1} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \frac{p}{p+1} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \\
&+ \frac{1}{2} \sum_{i=2}^p \left[\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla^i F(x) - D^{(i)}(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \|y-x\|^{\frac{p+1}{p}} + \frac{M \cdot \|y-x\|^{p+1}}{2p \cdot (p+1)!}\right] \\
&< -\frac{M}{8(p+1)!} \|y-x\|^{p+1} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \frac{p}{p+1} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \\
&+ \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla^i F(x) - D^{(i)}(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \eta^{\frac{p+1}{p}} \\
&< -\frac{M}{8(p+1)!} \|y-x\|^{p+1} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} + \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla^i F(x) - D^{(i)}(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \eta^{\frac{p+1}{p}}
\end{aligned}$$

378 which finishes the proof. $\square$

379 **Lemma 4.** Given a function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, let $y \in \operatorname{argmin}_{z: \|z-x\| \leq \eta} m_x(z)$, where

$$m_x(y) = F(x) + \langle D^{(1)}, y-x \rangle + \sum_{i=2}^p \frac{1}{i!} D^{(i)}[y-x]^i + \frac{M}{(p+1)!} \|y-x\|^{p+1}$$

380 for $M \geq 8L_p$ and $0 \leq \eta < 1$. It holds that:

$$\mathbf{1}[\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p] \leq \frac{1}{\eta^p} \|y-x\|^p + \frac{p!}{M\eta^p} (\|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\|_{\text{op}} \cdot \eta^i)$$

381 *Proof.* We have that

$$\begin{aligned}
&\|\nabla F(y)\| \\
&\leq \|\nabla F(y) - \sum_{i=0}^{p-1} \frac{1}{i!} \nabla^{i+1} F(x)[y-x]^i\| + \|\sum_{i=0}^{p-1} \frac{1}{i!} \nabla^{i+1} F(x)[y-x]^i\| \\
&\leq \frac{L_p}{p!} \|y-x\|^p + \|\nabla F(x) + \sum_{i=1}^{p-1} \frac{1}{i!} \nabla^{i+1} F(x)[y-x]^i\| \\
&\leq \frac{L_p}{p!} \|y-x\|^p + \|\nabla F(x) - D^{(1)}(x)\| + \|\sum_{i=1}^{p-1} \frac{1}{i!} [\nabla^{i+1} F(x)[y-x]^i - D^{(i+1)}(x)[y-x]^i]\| \\
&+ \|D^{(1)}(x) + \sum_{i=1}^{p-1} \frac{1}{i!} D^{(i+1)}[y-x]^i\| \\
&\leq \frac{L_p}{p!} \|y-x\|^p + \|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \|y-x\|^i \\
&+ \|D^{(1)}(x) + \sum_{i=1}^{p-1} \frac{1}{i!} D^{(i+1)}[y-x]^i\| \\
&\leq \frac{L_p + M}{p!} \|y-x\|^p + \|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \|y-x\|^i \\
&\leq \frac{L_p + M}{p!} \|y-x\|^p + \|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i
\end{aligned}$$

382 since under first order optimality conditions for $y \in \operatorname{argmin}_z m_x(z)$, we have that

$$D^{(1)}(x) + \sum_{i=1}^{p-1} \frac{1}{i!} D^{(i+1)}[y-x]^i + \frac{M}{p!} \|y-x\|^{p+1} (x-y) = 0$$

383 We now have that

$$\begin{aligned}
\|y - x\|^p &\geq \frac{p!}{L_p + M} (\|\nabla F(y)\| - \|\nabla F(x) - D^{(1)}(x)\| - \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i) \\
&\geq \min\{\eta^p, \frac{p!}{L_p + M} (\|\nabla F(y)\| - \|\nabla F(x) - D^{(1)}(x)\| - \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i)\} \\
&\geq \min\{\eta^p, \frac{p!}{L_p + M} \|\nabla F(y)\| - \frac{p!}{L_p + M} \|\nabla F(x) - D^{(1)}(x)\| - \frac{p!}{L_p + M} \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i\}
\end{aligned}$$

384 and since $L_p \leq \frac{M}{8}$ and $\frac{M}{L_p + M} < 1$, we have that

$$\begin{aligned}
M\|y - x\|^p &\geq \min\{M\eta^p, \frac{Mp!}{L_p + M} \|\nabla F(y)\| - \frac{Mp!}{L_p + M} \|\nabla F(x) - D^{(1)}(x)\| \\
&\quad - \frac{Mp!}{L_p + M} \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i \\
&> \min\{M\eta^p, \frac{8p!}{9} \|\nabla F(y)\| - p! (\|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i)\}
\end{aligned}$$

385 which means that

$$\min\{M\eta^p, \frac{8p!}{9} \|\nabla F(y)\|\} < M\|y - x\|^p + p! (\|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i)$$

386 which then implies that (since for all $a, b \geq 0$, $a\mathbf{1}[b \geq a] \leq \min\{a, b\}$)

$$\begin{aligned}
M\eta^p \cdot \mathbf{1}[\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p] &\leq M\|y - x\|^p + p! (\|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i) \\
\implies \mathbf{1}[\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p] &\leq \frac{1}{\eta^p} \|y - x\|^p + \frac{p!}{M\eta^p} (\|\nabla F(x) - D^{(1)}(x)\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{(i+1)}(x)\| \cdot \eta^i)
\end{aligned}$$

387 which proves the lemma. $\square$

388 **Lemma 5.** We now consider the setting where the derivative estimates D^i are random variables. For
389 all $F \in \mathcal{F}_p(\Delta, L_{1,p})$, it holds that

$$\begin{aligned}
\mathbb{E}[F(x) - F(y)] &\geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p+1}} \right] \cdot \left[\left(\frac{\sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
&\quad - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p+1}} \right) \sum_{i=2}^p \left[\left(\frac{\sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]
\end{aligned}$$

390 *Proof.* First, we note that

$$\begin{aligned}
\mathbf{1}[\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p] &\leq \left(\frac{1}{\eta^p} \|y - x\|^p + \frac{p!}{M\eta^p} (\|\nabla F(x) - g\| + \sum_{i=1}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{i+1}(x)\| \cdot \eta^i) \right)^{\frac{p+1}{p}} \\
&\leq \frac{2^{1/p}}{\eta^{p+1}} \|y - x\|^{p+1} + 2^{1/p} \cdot \left(\frac{p!}{M\eta^p} \right)^{\frac{p+1}{p}} \cdot \left(\sum_{i=0}^{p-1} \frac{1}{i!} \|\nabla^{i+1} F(x) - D^{i+1}(x)\| \cdot \eta^i \right)^{\frac{p+1}{p}} \\
&< \frac{2}{\eta^{p+1}} \|y - x\|^{p+1} + \frac{2(p!)^{\frac{p+1}{p}}}{M^{\frac{p+1}{p}} \eta^{p+1}} \cdot \left(\sum_{i=0}^{p-1} \|\nabla^{i+1} F(x) - D^{i+1}(x)\| \cdot \eta^i \right)^{\frac{p+1}{p}}
\end{aligned}$$

391 where we used the fact that for any $a_i \geq 0$, we have that

$$\left(\sum_{i=1}^n a_i\right)^{\frac{p+1}{p}} \leq n^{\frac{1}{p}} \sum_{i=1}^n a_i^{\frac{p+1}{p}}$$

392 which follows from an application of Hölder's inequality. We can now continue to bound the above
393 expression by

$$\begin{aligned} & \frac{2}{\eta^{p+1}} \|y - x\|^{p+1} + \frac{2(p!)^{\frac{p+1}{p}}}{M^{\frac{p+1}{p}} \eta^{p+1}} \cdot p^{1/p} \cdot \sum_{i=0}^{p-1} \|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}} \\ & < \frac{2}{\eta^{p+1}} \|y - x\|^{p+1} + \frac{4(p!)^{\frac{p+1}{p}}}{M^{\frac{p+1}{p}} \eta^{p+1}} \cdot \sum_{i=0}^{p-1} \|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}} \end{aligned}$$

394 where we used the fact that for all $p \geq 1$, $p^{1/p} < 2$. Taking expectations on each side, we have that

$$\mathbb{E}[\|y - x\|^{p+1}] \geq \frac{\eta^{p+1}}{2} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \frac{2(p!)^{\frac{p+1}{p}}}{M^{\frac{p+1}{p}}} \cdot \sum_{i=0}^{p-1} \mathbb{E}[\|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}}]$$

395 We also have that

$$\begin{aligned} & \mathbb{E}[F(x) - F(y)] \\ & > \frac{M}{8(p+1)!} \|y - x\|^{p+1} - \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}} \\ & - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \|\nabla^i F(x) - D^i(x)\|_{\text{op}}^{\frac{p+1}{p}} \cdot \eta^{\frac{p+1}{p}} \\ & \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \frac{2(p!)^{\frac{p+1}{p}}}{\sqrt[p]{M} \cdot 8(p+1)!} \cdot \sum_{i=0}^{p-1} \mathbb{E}[\|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}}] \\ & - \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}}] - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \cdot \eta^{\frac{p+1}{p}} \\ & \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} \cdot \sum_{i=0}^{p-1} \mathbb{E}[\|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}}] \\ & - \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}}] - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \cdot \eta^{\frac{p+1}{p}} \\ & = \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot \mathbb{E}[\|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}}] \\ & - \frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} \cdot \sum_{i=1}^{p-1} \mathbb{E}[\|\nabla^{i+1} F(x) - D^{i+1}(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{i(p+1)}{p}}] - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \cdot \eta^{\frac{p+1}{p}} \\ & = \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot \mathbb{E}[\|\nabla F(x) - D^{(1)}(x)\|^{\frac{p+1}{p}}] \\ & - \frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} \cdot \sum_{i=2}^p \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{(i-1)(p+1)}{p}}] - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \cdot \eta^{\frac{p+1}{p}} \\ & \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot \mathbb{E}[\|\nabla F(x) - g\|^{\frac{p+1}{p}}] \\ & - \frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} \cdot \sum_{i=2}^p \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}} \cdot \eta^{\frac{(p+1)}{p}}] - \frac{1}{2} \sum_{i=2}^p \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \cdot \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \cdot \eta^{\frac{p+1}{p}} \end{aligned}$$

396 where the last step follows from the fact that $\eta \leq 1$. We further lower bound this expression by

$$\begin{aligned}
& \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \mathbb{E}[\|\nabla F(x) - g\|^{\frac{p+1}{p}}] \\
& - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2} \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \mathbb{E}[\|\nabla^i F(x) - D^i(x)\|^{\frac{p+1}{p}}] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot 2^{1/p} \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2} \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p 2^{1/p} \cdot \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(y)\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]
\end{aligned}$$

397 where we used the fact that $2^{1/p} \leq 2$ for all $p \geq 1$, finishing the proof. $\square$

398 **Lemma 6.** Let $F \in \mathcal{F}_p(\Delta, L_{1,p})$ be given. Then, if the derivative estimates D^i are random variables,
399 it holds that

$$\begin{aligned}
& \Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!} \eta^p) \leq \frac{16(p+1)!}{M\eta^{p+1}T} \Delta + \frac{16(p+1)!}{M\eta^{p+1}} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& + \frac{16(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]
\end{aligned}$$

400 *Proof.* From lemma 5, we have that

$$\begin{aligned}
& \mathbb{E}[F(x^{(t)}) - F(x^{(t+1)})] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(x^{(t+1)})\| \geq \frac{9M}{8p!} \eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]
\end{aligned}$$

401 Telescoping this recurrence from $t = 1$ to T gives

$$\begin{aligned}
& \mathbb{E}[F(x^{(1)}) - F(x^{(T+1)})] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \cdot T \cdot \left(\frac{1}{T} \sum_{t=1}^T \Pr(\|\nabla F(x^{(t+1)})\| \geq \frac{9M}{8p!} \eta^p) \right) - T \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& - T \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right] \\
& = \frac{M\eta^{p+1}}{16(p+1)!} \cdot T \cdot (\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!} \eta^p)) - T \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \\
& - T \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]
\end{aligned}$$

402 which implies that

$$\begin{aligned}
\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p) &\leq \frac{16(p+1)!}{M\eta^{p+1}T}\Delta + \frac{16(p+1)!}{M\eta^{p+1}} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}}\right] \right. \\
&+ \left. \frac{16(p+1)!}{M\eta^{p+1}} \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right] \right] \\
&= \frac{16(p+1)!}{M\eta^{p+1}T}\Delta + \frac{16(p+1)!}{M\eta^{p+1}} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}}\right] \right. \\
&+ \left. \frac{16(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right] \right]
\end{aligned}$$

403 which finishes the proof. $\square$

404 **Theorem 5.** (Theorem 3 restated). For any function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, where $p \geq 2$, $\epsilon > 0$,
405 with biased and stochastic p^{th} -order oracles in $\mathcal{O}(F, \sigma_{1:p}, B_{1:p})$ where $\max_i B_i \geq \Omega(\epsilon^{\frac{3p}{3p+1}})$, with
406 probability at least $\frac{5}{8}$, Algorithm 1 returns a point $\hat{x}$ such that $\|\nabla F(\hat{x})\| \leq O(\epsilon + \max_i B_i)$ and
407 performs at most

$$O\left(\frac{\Delta(\max_i \sigma_i)^2}{\epsilon^3(\max_i B_i + 1)^{\frac{p+1}{p}}} + \frac{(\epsilon + \max_i B_i)^{\frac{p+1}{p}}(\max_i \sigma_i)^2}{\epsilon^3(\max_i B_i + 1)^{\frac{p+1}{p}}}\right)$$

408 queries to the stochastic and biased derivative oracles.

409 *Proof.* From lemma 6, we have that

$$\begin{aligned}
\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p) &\leq \frac{16(p+1)!}{M\eta^{p+1}T}\Delta + \frac{16(p+1)!}{M\eta^{p+1}} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \cdot \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}}\right] \right. \\
&+ \left. \frac{16(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \left(\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right) \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right] \right]
\end{aligned}$$

410 Let

$$A = \max\left(\frac{16(p+1)!}{M}, \frac{16(p+1)!}{M} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + 2\left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right], \frac{16(p+1)!}{M} \left[\frac{(p!)^{\frac{p+1}{p}}}{4\sqrt[p]{M}} + \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right]\right)$$

411 Therefore, we have the following upper bound:

$$\frac{A\Delta}{\eta^{p+1}T} + \frac{A}{\eta^{p+1}} \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] + \frac{A}{\eta^{\frac{p^2-1}{p}}} \sum_{i=2}^p \left[\left(\frac{C_i \cdot \sigma_i^2}{n_i}\right)^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}} \right]$$

412 From our choice of n_1 (where such an n_1 exists due to $\max_i B_i \geq \Omega(\epsilon^{\frac{3p}{3p+1}})$) such that

$$\max\left\{\frac{C_1 \cdot \sigma_1^2}{\left(\frac{\eta^{p+1}}{8A} - B_1^{\frac{p+1}{2p}}\right)^{\frac{2p}{p+1}}}, 1\right\} \leq n_1 \leq \frac{(\epsilon + \max_i B_i)^{\frac{p+1}{p}}(\max_i \sigma_i)^2}{\epsilon^3(\max_i B_i + 1)^{\frac{p+1}{p}}}$$

413 we have that

$$\frac{A}{\eta^{p+1}} \left[\left(\frac{C_1 \cdot \sigma_1^2}{n_1}\right)^{\frac{p+1}{2p}} + B_1^{\frac{p+1}{2p}} \right] \leq \frac{1}{8}$$

414 From our choice of $T = \lceil \frac{8A\Delta}{\eta^{p+1}} \rceil$ we have that

$$\frac{A\Delta}{\eta^{p+1}T} \leq \frac{1}{8}$$

415 From our choice of n_i (for all $i \geq 2$, which exists due to $\max_i B_i \geq \Omega(\epsilon^{\frac{3p}{3p+1}})$) such that

$$\max\{C_i \sigma_i^2 (\frac{\eta^{\frac{p^2-1}{p}}}{8Ap} - B_i^{\frac{p+1}{2p}})^{\frac{-2p}{p+1}}, 1\} \leq n_i \leq \frac{(\epsilon + \max_i B_i)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (\max_i B_i + 1)^{\frac{p+1}{p}}}$$

416 it holds that

$$\frac{A}{\eta^{\frac{p^2-1}{p}}} [(\frac{C_i \cdot \sigma_i^2}{n_i})^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}}] \leq \frac{1}{8p}$$

417 which implies that

$$\frac{A}{\eta^{\frac{p^2-1}{p}}} \sum_{i=2}^p [(\frac{C_i \cdot \sigma_i^2}{n_i})^{\frac{p+1}{2p}} + B_i^{\frac{p+1}{2p}}] \leq \frac{(p-1)}{8p} \leq \frac{1}{8}$$

418 Therefore, we have that

$$\Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!} (\epsilon + \max_i B_i)) \leq \frac{3}{8}$$

419 which implies that

$$\Pr(\|\nabla F(\hat{x})\| < \frac{9M}{8p!} (\epsilon + \max_i B_i)) \geq \frac{5}{8}$$

420 Now, we give a bound on the oracle complexity. Let M be the total number of oracle queries that we
421 make. In every iteration, we query the i^{th} derivative oracle n_i times which yields that

$$\begin{aligned} \mathbb{E}[M] &= T \sum_{i=1}^p n_i \\ &\leq (\frac{8A\Delta}{\eta^{p+1}} + 1) \sum_{i=1}^p n_i \end{aligned}$$

422 Substituting the upper bound for n_i , we can further bound this expression by

$$\begin{aligned} &(\frac{8A\Delta}{\eta^{p+1}} + 1) \sum_{i=1}^p \frac{(\epsilon + \max_j B_j)^{\frac{p+1}{p}} \cdot (\max_i \sigma_i)^2}{\epsilon^3 (\max_j B_j + 1)^{\frac{p+1}{p}}} \\ &\leq O(\frac{\Delta}{(\epsilon + \max_j B_j)^{\frac{p+1}{p}}} \cdot \frac{(\epsilon + \max_j B_j)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (\max_j B_j + 1)^{\frac{p+1}{p}}} + \frac{(\epsilon + \max_j B_j)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (\max_j B_j + 1)^{\frac{p+1}{p}}}) \\ &= O(\frac{\Delta (\max_i \sigma_i)^2}{\epsilon^3 (\max_j B_j + 1)^{\frac{p+1}{p}}} + \frac{(\epsilon + \max_j B_j)^{\frac{p+1}{p}} (\max_i \sigma_i)^2}{\epsilon^3 (\max_j B_j + 1)^{\frac{p+1}{p}}}) \end{aligned}$$

423 which finishes the proof. $\square$

424 B Appendix B

425 **Lemma 7.** Let $F \in \mathcal{F}_p(\Delta, L_{1:p})$. For any biased and stochastic oracle in $\mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$,
 426 let $\{D^i(x^{(t)})\}$ represent the sequence of i^{th} derivative iterates generated by Algorithm 2. Let
 427 $B = \max_{1 \leq i \leq p} B_i$. Then, we have that

$$\mathbb{E}[\|D^i(x^{(t)}) - \nabla^i F(x^{(t)})\|^2] \leq 4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p}$$

428 for all $1 \leq i \leq p$ and all $t \geq 1$.

429 *Proof.* We can first say that

$$\begin{aligned} & \mathbb{E}[\|D^i(x^{(1)}) - \nabla^i F(x^{(1)})\|^2] \\ &= \mathbb{E}[\|b_i(x^{(1)}) + \frac{1}{n_1} \sum_{j=1}^{n_1} \epsilon_1(x^{(1)}, z^{(1,j)})\|^2] \\ &\leq 2B_i^2 + \frac{2}{n_i^2} \sum_{j=1}^{n_i} \|\epsilon_i(x^{(1)}, z^{(1,j)})\|^2 \leq 2B_i^2 + \frac{2\sigma_i^2}{n_i} \leq 2B_i^2 + \frac{2\epsilon^2}{5} \end{aligned}$$

430 Let $e^{(t)} = D_t^i(x^{(t)}) - \nabla F^i(x^{(t)})$, and we have that

$$\mathbb{E}[\|e^{(t)}\|^2 | b^{(t)}] = b^{(t)} \cdot \mathbb{E}[\|e^{(t)}\|^2 | C^{(t)} = 1] + (1 - b^{(t)}) \cdot \mathbb{E}[\|e^{(t)}\|^2 | C^{(t)} = 0]$$

431 where

$$\mathbb{E}[\|e^{(t)}\|^2 | C^{(t)} = 1] \leq 2B_i^2 + \frac{2\sigma_i^2}{n_i} \leq 2B_i^2 + \frac{2\epsilon^2}{5}$$

432 We now say that

$$\begin{aligned} & \mathbb{E}[\|e^{(t)}\|^2 | C^{(t)} = 0] \\ &\leq \mathbb{E}[\|e^{(t-1)}\|^2] + \mathbb{E}[\|\psi^{(t)} | \mathcal{G}^{(t)}\|^2] + \mathbb{E}[\|\psi^{(t)} - \mathbb{E}[\psi^{(t)} | \mathcal{G}^{(t)}]\|^2] \\ &\leq \mathbb{E}[(1 + \frac{2}{b^{(t)}}) \cdot \|e^{(t-1)}\|^2] + \mathbb{E}[(1 + \frac{2}{b^{(t)}}) \cdot \|\mathbb{E}[\psi^{(t)} | \mathcal{G}^{(t)}]\|^2] + \mathbb{E}[\|\psi^{(t)} - \mathbb{E}[\psi^{(t)} | \mathcal{G}^{(t)}]\|^2] \end{aligned}$$

433 where the first step follows from the fact that $\mathcal{G}^{(t)}$ is a measurable set, and the second step is by

434 Young's inequality. Above, we have that

$$\psi^{(t)} = e^{(t)} - e^{(t-1)} = \sum_{k=1}^{K^{(t)}} \tilde{\nabla}^{i+1} F(x^{(t,k-1)}, z^{(t,k)}, b_i)(x^{(t,k)} - x^{(t,k-1)}) - \nabla^i F(x^{(t)}) + \nabla^i F(x^{(t-1)})$$

435 We can calculate that

$$\mathbb{E}[\psi^{(t)} | \mathcal{G}^{(t)}] = \sum_{k=1}^{K^{(t)}} (\nabla^{i+1} F(x^{(t,k-1)}) + b_{i+1}(x^{(t,k-1)}))(x^{(t,k)} - x^{(t,k-1)}) - \nabla^i F(x^{(t)}) + \nabla^i F(x^{(t-1)})$$

436 which implies that

$$\begin{aligned} & \|\mathbb{E}[\psi^{(t)} | \mathcal{G}^{(t)}]\| \\ &\leq \sum_{k=1}^{K^{(t)}} \|(\nabla^i F(x^{(t,k)}) - \nabla^i F(x^{(t,k-1)}) - \nabla^{i+1} F(x^{(t,k-1)})(x^{(t,k)} - x^{(t,k-1)}))\| \\ &+ \sum_{k=1}^{K^{(t)}} \|b_{i+1}(x^{(t,k-1)})\| \cdot \|x^{(t,k)} - x^{(t,k-1)}\| \\ &\leq K^{(t)} \cdot \frac{L_{i+1}}{2} \cdot (\frac{\|x^{(t)} - x^{(t-1)}\|}{K^{(t)}})^2 + B_{i+1} \|x^{(t)} - x^{(t-1)}\| \\ &\leq \frac{b^{(t)}\epsilon}{10} + B_{i+1}\eta \\ &\leq \frac{b^{(t)}\epsilon}{10} + B_{i+1}b^{(t)}(\epsilon + \max_i B_i)^{1/p} \end{aligned}$$

437 We can also derive that

$$\begin{aligned}
& \mathbb{E}[\|\psi^{(t)} - \mathbb{E}[\psi^{(t)}|\mathcal{G}^{(t)}]\|^2] \\
&= \frac{1}{(K^{(t)})^2} \sum_{k=1}^{K^{(t)}} \mathbb{E}[\|(\tilde{\nabla}^{i+1}F(x^{(t,k-1)}, z^{(t,k)}) - \nabla^{i+1}F(x^{(t,k-1)}) - b_{i+1}(x^{(t,k-1)})(x^{(t)} - x^{(t-1)}))\|^2|\mathcal{G}^{(t)}] \\
&\leq \frac{1}{(K^{(t)})^2} \sum_{k=1}^{K^{(t)}} \mathbb{E}[\|(\tilde{\nabla}^{i+1}F(x^{(t,k-1)}, z^{(t,k)}) - \nabla^{i+1}F(x^{(t,k-1)}) - b_{i+1}(x^{(t,k-1)}))\|_{\text{op}}^2|\mathcal{G}^{(t)}] \cdot \|x^{(t)} - x^{(t-1)}\|^2 \\
&\leq \sigma_{i+1}^2 \frac{\|x^{(t)} - x^{(t-1)}\|^2}{K^{(t)}} \leq b^{(t)} \frac{\epsilon^2}{5}
\end{aligned}$$

438 Combining both of these inequalities together, we have that

$$\begin{aligned}
& \mathbb{E}\|e^{(t)}\|^2 \\
&= b^{(t)}(2B_i^2 + \frac{2\epsilon^2}{5}) + (1 - b^{(t)}) \cdot \mathbb{E}[\|e^{(t)}\|^2|C^{(t)} = 0] \\
&\leq b^{(t)}(2B_i^2 + \frac{2\epsilon^2}{5}) + \mathbb{E}[(1 - b^{(t)})(1 + \frac{2}{b^{(t)}})\|e^{(t-1)}\|^2 + (1 - b^{(t)})(1 + \frac{2}{b^{(t)}})(\frac{b^{(t)}\epsilon}{10} + B_{i+1}b^{(t)}(\epsilon + \max_i B_i)^{\frac{1}{p}})^2] \\
&+ \mathbb{E}[(1 - b^{(t)}) \cdot \frac{b^{(t)}\epsilon^2}{5}] \\
&\leq \mathbb{E}[b^{(t)} \cdot (2B^2 + \frac{2\epsilon^2}{5})] + (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \mathbb{E}[3b^{(t)} \cdot (\epsilon + B(\epsilon + B)^{1/p})^2 + b^{(t)}\frac{\epsilon^2}{5}] \\
&\leq (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \mathbb{E}[b^{(t)}](2B^2 + \frac{3\epsilon^2}{5} + 3(\epsilon + B(\epsilon + B)^{1/p})^2) \\
&\leq (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \mathbb{E}[b^{(t)}] \cdot (2B^2 + \frac{3\epsilon^2}{5} + 3(\epsilon + B\epsilon^{1/p} + B^{1/p})^2) \\
&\leq (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \mathbb{E}[b^{(t)}] \cdot (2B^2 + \frac{3\epsilon^2}{5} + 9\epsilon^2 + 9B^2\epsilon^{2/p} + 9B^{2/p}) \\
&= (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \mathbb{E}[b^{(t)}] \cdot (2B^2 + \frac{48\epsilon^2}{5} + 9B^2\epsilon^{2/p} + 9B^{2/p}) \\
&= (1 - \frac{\mathbb{E}[b^{(t)}]}{2})\|e^{(t-1)}\|^2 + \frac{\mathbb{E}[b^{(t)}]}{2} \cdot (4B^2 + \frac{96\epsilon^2}{5} + 18B^2\epsilon^{2/p} + 18B^{2/p})
\end{aligned}$$

439 which implies that

$$\begin{aligned}
& \mathbb{E}\|e^{(t)}\|^2 \\
&\leq (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p}) - (2B^2 + \frac{94}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p}) \prod_{s=2}^t (1 - \frac{b^{(s)}}{2}) \\
&\leq 4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p}
\end{aligned}$$

440 which finishes the proof. $\square$

441 **Lemma 8.** *It holds that*

$$\begin{aligned}
& \Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p) \leq \frac{16(p+1)!}{M\eta^{p+1}T}\Delta + \frac{16(p+1)!}{M\eta^{p+1}} \cdot [\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + (\frac{2p!}{M})^{\frac{1}{p}}] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
&+ \frac{16p(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \cdot [\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}(\frac{2p \cdot p!}{M})^{\frac{1}{p}}] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}}
\end{aligned}$$

442 *Proof.* From Lemma 5, we have that

$$\begin{aligned}
& \mathbb{E}[F(x^{(t)}) - F(x^{(t+1)})] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(x^{(t+1)})\| \geq \frac{9M}{8p!}\eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot \mathbb{E}[\|\nabla F(x^{(t)}) - D^{(1)}(x^{(t)})\|^{\frac{p+1}{p}}] \\
& \quad - \eta^{\frac{p+1}{p}} \left(\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}}\right) \sum_{i=2}^p \mathbb{E}[\|\nabla^i F(x^{(t)}) - D^{(i)}(x^{(t)})\|^{\frac{p+1}{p}}] \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(x^{(t+1)})\| \geq \frac{9M}{8p!}\eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot (\mathbb{E}[\|\nabla F(x^{(t)}) - D^{(1)}(x^{(t)})\|^2])^{\frac{p+1}{2p}} \\
& \quad - \eta^{\frac{p+1}{p}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}}\right] \cdot \sum_{i=2}^p (\mathbb{E}[\|\nabla^i F(x^{(t)}) - D^{(i)}(x^{(t)})\|^2])^{\frac{p+1}{2p}} \\
& \geq \frac{M\eta^{p+1}}{16(p+1)!} \Pr(\|\nabla F(x^{(t+1)})\| \geq \frac{9M}{8p!}\eta^p) - \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{\frac{2}{p}} + 18B^{\frac{2}{p}})^{\frac{p+1}{2p}} \\
& \quad - p\eta^{\frac{p+1}{p}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{\frac{2}{p}} + 18B^{\frac{2}{p}})^{\frac{p+1}{2p}}
\end{aligned}$$

443 Telescoping this recurrence from $t = 1$ to T gives

$$\begin{aligned}
& \mathbb{E}[F(x^{(1)}) - F(x^{(T+1)})] \\
& \geq \frac{M\eta^{p+1}T}{16(p+1)!} \Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p) - T \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& \quad - p\eta^{\frac{p+1}{p}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}}
\end{aligned}$$

444 which implies that

$$\begin{aligned}
& \Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!}\eta^p) \leq \frac{16(p+1)!}{M\eta^{p+1}T} \Delta + \frac{16(p+1)!}{M\eta^{p+1}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& \quad + \frac{16p(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2}\left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}}\right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}}
\end{aligned}$$

445 which finishes the proof. $\square$

446 **Theorem 6.** *Theorem (4 restated). For any function $F \in \mathcal{F}_p(\Delta, L_{1:p})$, with biased and stochastic*
447 *p^{th} order oracles in $\mathcal{O}(F, \sigma_{1:p}, B_{1:p})$, with probability at least $\frac{5}{8}$, Algorithm 3 returns a point $\hat{x}$ such*
448 *that:*

449 • If $\max_i B_i = \Theta(1)$, then $\|\nabla F(\hat{x})\| \leq O(\epsilon + \max_i B_i)$ with at most

$$O\left(\frac{\Delta(\max_i \sigma_i)^2(\epsilon + B)^{\frac{1}{p}} + (\max_i \sigma_i)^2}{\epsilon^2} + \frac{\Delta(\epsilon + B)^{\frac{1}{p}} + 1}{\epsilon}\right)$$

450 queries to the stochastic and biased derivative oracles.

451 • If $\max_i B_i > \Omega(1)$, then $\|\nabla F(\hat{x})\| \leq O((\epsilon^2 + B^2)^{\frac{1}{2}}(\epsilon + B))$ with at most

$$O\left(\frac{(\max_i \sigma_i)^2}{\epsilon^2(\epsilon + B)^{\frac{p+1}{p}}} + \frac{1}{\epsilon(\epsilon + B)^{\frac{p+1}{p}}}\right)$$

452 queries to the stochastic and biased derivative oracles.

453 where $B = \max_i B_i$.

454 *Proof.* From Lemma 8, we have that

$$\begin{aligned}
& \Pr(\|\nabla F(\hat{x})\| \geq \frac{9M}{8p!} \eta^p) \\
& \leq \frac{16(p+1)!}{M\eta^{p+1}T} \Delta + \frac{16(p+1)!}{M\eta^{p+1}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \left(\frac{2p!}{M}\right)^{\frac{1}{p}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& + \frac{16p(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}}}{8\sqrt[p]{M}} + \frac{1}{2} \left(\frac{2p \cdot p!}{M}\right)^{\frac{1}{p}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& = \frac{16(p+1)!}{M\eta^{p+1}T} \Delta + \frac{16(p+1)!}{M\eta^{p+1}} \left[\frac{(p!)^{\frac{p+1}{p}} + 8 \cdot (2p!)^{\frac{1}{p}}}{8M^{\frac{1}{p}}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& + \frac{16p(p+1)!}{M\eta^{\frac{p^2-1}{p}}} \cdot \left[\frac{(p!)^{\frac{p+1}{p}} + 4(2p \cdot p!)^{\frac{1}{p}}}{8M^{\frac{1}{p}}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& = \frac{16(p+1)!}{M\eta^{p+1}T} \Delta + \frac{2(p+1)!}{M^{\frac{p+1}{p}} \eta^{p+1}} \left[(p!)^{\frac{p+1}{p}} + 8 \cdot (2p!)^{\frac{1}{p}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& + \frac{2p(p+1)!}{M^{\frac{p+1}{p}} \eta^{\frac{p^2-1}{p}}} \cdot \left[(p!)^{\frac{p+1}{p}} + 4(2p \cdot p!)^{\frac{1}{p}} \right] \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}}
\end{aligned}$$

455 Letting

$$A = \max(16(p+1)!, 2(p+1)! \cdot [(p!)^{\frac{p+1}{p}} + 8 \cdot (2p!)^{\frac{1}{p}}], 2(p+1)! \cdot [(p!)^{\frac{p+1}{p}} + 4(2p \cdot p!)^{\frac{1}{p}}])$$

456 we get an upper bound of

$$\begin{aligned}
& \frac{A\Delta}{M\eta^{p+1}T} + \frac{A}{M^{\frac{p+1}{p}} \eta^{p+1}} \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \\
& + \frac{A}{M^{\frac{p+1}{p}} \eta^{\frac{p^2-1}{p}}} \cdot (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}}
\end{aligned}$$

457 Let $X = 4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p}$. Since we set

$$M = \max\left\{\left(\frac{8AX^{\frac{p+1}{2p}}}{\eta^{p+1}}\right)^{\frac{p}{p+1}}, \left(\frac{8Ap \cdot X^{\frac{p+1}{2p}}}{\eta^{\frac{p^2-1}{p}}}\right)^{\frac{p}{p+1}}, (\epsilon + B)^{\frac{-p-2}{p}}, 8L_p\right\}$$

458 we have that

$$\frac{A}{M^{\frac{p+1}{p}} \eta^{p+1}} (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \leq \frac{1}{8}$$

459 and

$$\frac{pA}{M^{\frac{p+1}{p}} \eta^{\frac{p^2-1}{p}}} (4B^2 + \frac{96}{5}\epsilon^2 + 18B^2\epsilon^{2/p} + 18B^{2/p})^{\frac{p+1}{2p}} \leq \frac{1}{8}$$

460 and through setting

$$T = \lceil \frac{8A\Delta}{M\eta^{p+1}} \rceil > \frac{8A\Delta}{M\eta^{p+1}}$$

461 we have that

$$\frac{A\Delta}{M\eta^{p+1}T} < \frac{1}{8}$$

462 Notice that if $B = \Theta(1)$, then $X = \Theta(1)$, we have that

$$\frac{5}{8} \leq \Pr(\|\nabla F(\hat{x})\| \leq \frac{9M}{8p!} \eta^p) \leq \Pr(\|\nabla F(\hat{x})\| \leq O(1) \cdot (\epsilon + B))$$

On the other hand, if $B > \Omega(1)$, then $X = O(B^2 + \epsilon^2)$, which implies that $M = \max(O(\frac{(B^2 + \epsilon^2)^{1/2}}{\eta^p}), O(\frac{(B^2 + \epsilon^2)^{\frac{1}{2}}}{\eta^{p-1}}), O(1), O(1))$. Since $B > \Omega(1)$, $1 - \epsilon < (\epsilon + B)^{\frac{1}{p}}$, so $\eta = 1 - \epsilon$. Therefore, $M = O((B^2 + \epsilon^2)^{\frac{1}{2}})$. We then have that

$$\frac{5}{8} \leq \Pr(\|\nabla F(\hat{x})\| \leq \frac{9M}{8p!} \eta^p) \leq \Pr(\|\nabla F(\hat{x})\| \leq O((\epsilon^2 + B^2)^{\frac{1}{2}}(\epsilon + B)))$$

Let M_i be the number of oracle queries for derivative order i , and let $M = \sum_{i=1}^{p+1} M_i$ be the total number of oracle queries. With regards to the oracle complexity, we have that

$$\begin{aligned} \mathbb{E}[M] &\leq \sum_{i=1}^{p+1} \mathbb{E}[M_i] \\ &= T \sum_{i=1}^{p+1} \Pr(C = 1) \mathbb{E}[m_i | C = 1] + \Pr(C = 0) \mathbb{E}[m_i | C = 0] \\ &= T \sum_{i=1}^{p+1} b n_i + (1 - b) K_i \\ &\leq T \sum_{i=1}^{p+1} b \left(\frac{5\sigma_i^2}{\epsilon^2} + 1 \right) + (1 - b) \left(\frac{5(\sigma_{i+1}^2 + L_{i+1}\epsilon)}{b\epsilon^2} + 1 \right) \end{aligned}$$

Letting $\sigma = \max_i \sigma_i$, we upper bound the expression above by

$$\begin{aligned} &T \sum_{i=1}^{p+1} b \left(\frac{5\sigma^2}{\epsilon^2} + 1 \right) + (1 - b) \left(\frac{5(\sigma^2 + L_{i+1}\epsilon)}{b\epsilon^2} + 1 \right) \\ &\leq T \sum_{i=1}^{p+1} \frac{5b^2\sigma^2 + 5\sigma^2 + 5L_{i+1}\epsilon}{b\epsilon^2} + 2 \\ &\leq T \cdot O\left(\frac{\sigma^2}{\epsilon^2} + \frac{1}{\epsilon}\right) \end{aligned}$$

We again analyze the following cases: $B = \Theta(1)$ and $B > \Omega(1)$. If $B = \Theta(1)$, then

$$\frac{8A\Delta}{M\eta^{p+1}} \leq \frac{8A\Delta}{(\epsilon + B)^{-\frac{1}{p}} \eta^{p+1}} \leq \frac{8A\Delta}{(\epsilon + B)^{-\frac{p-2}{p}} (\epsilon + B)^{\frac{p+1}{p}} \cdot O(1)} = O(1) \cdot 8A\Delta \cdot (\epsilon + B)^{\frac{1}{p}}$$

Therefore,

$$T \cdot O\left(\frac{\sigma^2}{\epsilon^2} + \frac{1}{\epsilon}\right) \leq O\left(\frac{\Delta\sigma^2(\epsilon + B)^{\frac{1}{p}} + \sigma^2}{\epsilon^2} + \frac{\Delta(\epsilon + B)^{\frac{1}{p}} + 1}{\epsilon}\right)$$

If $B > \Omega(1)$, then

$$\frac{8A\Delta}{M\eta^{p+1}} = \frac{8A\Delta}{M(\epsilon + B)^{\frac{p+1}{p}}} \leq \frac{8A\Delta}{8L_p \cdot (\epsilon + B)^{\frac{p+1}{p}}} \leq O\left(\frac{\Delta}{L_p \cdot (\epsilon + B)^{\frac{p+1}{p}}}\right)$$

and so we have that

$$T \cdot O\left(\frac{\sigma^2}{\epsilon^2} + \frac{1}{\epsilon}\right) \leq O\left(\frac{\sigma^2}{\epsilon^2(\epsilon + B)^{\frac{p+1}{p}}} + \frac{1}{\epsilon(\epsilon + B)^{\frac{p+1}{p}}}\right)$$

which finishes the proof. $\square$

C Appendix C

C.1 Preliminaries

We first introduce some important notational conventions used throughout this section. Given a p^{th} order tensor $T \in \mathbb{R}^{d \times \dots \times d}$, we define the support of T as the following:

$$\text{supp}(T) = \{i \in [d] : T_i \neq 0\}$$

where T_i is the $(p-1)$ order subtensor denoted by $[T_i]_{j_1, \dots, j_{p-1}} = T_{i, j_1, \dots, j_{p-1}}$. For a tuple of tensors $\mathcal{T} = (T^{(1)}, T^{(2)}, \dots)$, we define

$$\text{supp}(\mathcal{T}) = \bigcup_i \text{supp}(T_i)$$

Moreover, given $x \in \mathbb{R}^d$, let

$$\text{prog}_\alpha(x) = \max\{i \geq 0, |x_i| > \alpha\}$$

which represents the highest index of x whose entry is at least α from zero. Notice that for any $\alpha_1, \alpha_2 \in [0, 1]$ such that $\alpha_1 < \alpha_2$, we have that $\text{prog}_{\alpha_2}(x) < \text{prog}_{\alpha_1}(x)$. For a tensor T , we define $\text{prog}(T) = \max\{\text{supp}\{T\}\}$ which represents the highest index in $\text{supp}\{T\}$, and naturally for a collection of tensors $\mathcal{T} = \{T^{(i)}\}$, we define $\text{prog}(\mathcal{T}) = \max_i \text{prog}(T^{(i)})$.

Definition 1. A biased and stochastic algorithm A is zero-respecting if for any function F and p th-order oracle O_F^p , the iterates $\{x^{(t)}\}$ satisfy

$$\text{supp}(x^{(t)}) \subseteq \bigcup_{i < t} \text{supp}(O_F^p(x^{(i)}, z^{(i)}, b^{(i)}))$$

for all $t \in \mathbb{N}$.

Definition 2. A collection of derivative estimators $\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)$ for a function F form a probability- ρ zero-chain if

$$\Pr(\exists x | \text{prog}(\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)) = \text{prog}_{\frac{1}{4}}(x) + 1) \leq \rho$$

and

$$\Pr(\exists x | \text{prog}(\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)) = \text{prog}_{\frac{1}{4}}(x) + i) = 0$$

for all $i > 1$.

Lemma 9. Let $\tilde{\nabla}^1 F(x, z, b), \dots, \tilde{\nabla}^p F(x, z, b)$ be a collection of probability- ρ zero-chain derivative estimators for $F : \mathbb{R}^T \rightarrow \mathbb{R}$, and let $O_F^p(x, z, b) = (\tilde{\nabla}^q F(x, z, b))_{q \in \{1, \dots, p\}}$. Let $\{x_{A[O_F]}^{(t)}\}$ be a sequence of queries produced by algorithm A interacting with O_F^p . Then, with probability at least $1 - \delta$,

$$\text{prog}(x^{(t)}) < T$$

for all

$$t \leq \frac{T - \log(1/\delta)}{2\rho}$$

Proof. Proved in Lemma 16 of [7] □

Definition 3. Let

$$F_T(x) = -\Psi(1)\Phi(1) + \sum_{i=2}^T [\Psi(-x_{i-1})\Phi(-x_i) - \Psi(x_{i-1})\Phi(x_i)]$$

where

$$\Psi(x) = \begin{cases} 0 & \text{if } x \leq \frac{1}{2} \\ \exp(1 - \frac{1}{(2x-1)^2}) & \text{if } x > \frac{1}{2} \end{cases}, \quad \Phi(x) = \sqrt{e} \int_{-\infty}^x e^{-\frac{1}{2}t^2} dt$$

Lemma 10. For F_T , the following properties hold:

- $F_T(0) - \inf_x F_T(x) \leq \Delta_0 T$, where $\Delta_0 = 12$
- For all $p \geq 1$, the p^{th} order derivatives of F_t are ℓ_p -Lipschitz continuous, where $\ell_p \leq \exp(\frac{5}{2}p \log p + cp)$ for some $c < \infty$

- 504 • For all $x \in \mathbb{R}^T$, $p \in \mathbb{N}$, and $1 \leq i \leq T$, we have that $\|\nabla_i^p F_T(x)\|_{\text{op}} \leq \ell_{p-1}$
- 505 • For all $x \in \mathbb{R}^T$, $p \in \mathbb{N}$, $\text{prog}(\nabla^q F_T(x)) \leq \text{prog}_{\frac{1}{2}}(x) + 1$
- 506 • For all $x \in \mathbb{R}^T$, if $\text{prog}_1(x) < T$, then $\|\nabla F_T(x)\| \geq |\nabla_{\text{prog}_1(x)+1} F_T(x)| > 1$
- 507 • For all $x, y \in \mathbb{R}^d$, there exists a constant $C \geq 0$ such that $\|F(x) - F(y)\| \leq C\sqrt{T}\|x - y\|$

508 *Proof.* The first five statements follow from Lemma 2, Lemma 3, and Observation 3 of [10] and
 509 section G.1.1. of [7]. We now prove the last statement coordinate-wise by considering three separate
 510 cases for a coordinate i : $1 < i < T$, $i = 1$, and $i = T$. First, let $1 < i < T$. So, we have that
 511 $\partial_i F_T(x)$

$$\begin{aligned} &= \frac{\partial}{\partial x_i} (\Psi(-x_{i-1})\Phi(-x_i) - \Psi(x_{i-1})\Phi(x_i)) + \frac{\partial}{\partial x_i} (\Psi(-x_i)\Phi(-x_{i+1}) - \Psi(x_i)\Phi(x_{i+1})) \\ &= -\Psi(-x_{i-1})\Phi'(-x_i) - \Psi(x_{i-1})\Phi'(x_i) - \Psi'(-x_i)\Phi(-x_{i+1}) - \Psi'(x_i)\Phi(x_{i+1}) \end{aligned}$$

512 Observe that by construction of Ψ , we have that $|\Psi(x)| \leq e$ and $|\Psi'(x)| \leq 5$. Also, observe that

$$\Phi(x) = \sqrt{e} \int_{-\infty}^x e^{-t^2/2} \leq \sqrt{e} \int_{-\infty}^{\infty} e^{-t^2/2} = \sqrt{2\pi e}$$

513 and that $|\Phi'(x)| \leq \sqrt{e}$. Therefore, it holds that

$$|\partial_i F_T(x)| \leq e\sqrt{e} + e\sqrt{e} + 5\sqrt{2\pi e} + 5\sqrt{2\pi e} \leq 51$$

514 One can derive similar constants for the $i = 1$ and $i = T$ case. Let C be the maximum of 51 and
 515 these constants. We have that

$$\|\nabla F_T(x)\|_2 = \left(\sum_{i=1}^T |\partial_i F_T(x)|^2 \right)^{\frac{1}{2}} \leq \left(\sum_{i=1}^T C^2 \right)^{\frac{1}{2}} = C\sqrt{T}$$

516 The statement follows by applying norm equivalence in finite dimensional spaces.

517 □

518 **Definition 4.** For all q , define the derivative estimators used to be

$$[\tilde{\nabla}^q F_T(x, z)]_i = (1 + \mathbf{1}\{i > \text{prog}_{\frac{1}{4}}(x)\}) \left(\frac{z}{\rho} - 1 \right) \cdot (\nabla_i^q F_T(x) + b_i^q(x))$$

519 where b^q is such that $b_i^q(x) = 0$ for all $i > \text{prog}_{1/4}(x) + 1$, $\|b_i^q(x)\| \leq B_q$, and $z \sim \text{Bernoulli}(\rho)$.

520 **Lemma 11.** The derivative estimators $\tilde{\nabla}^q F_T$ form a probability- ρ zero-chain and satisfy:

$$\mathbb{E}[\|\tilde{\nabla}^q F_T(x, z) - \nabla^q F_T(x)\|^2] \leq \frac{2\ell_{q-1}^2(1-\rho)}{\rho} + 2B_q^2$$

521 *Proof.* First, we prove that these derivative estimators form a probability- ρ chain. First, by the
 522 definition of F_T and b_i , we can immediately conclude that $[\tilde{\nabla}^q F_T(x, z)]_i = 0$ for all $i > \text{prog}_{\frac{1}{4}}(x) +$
 523 1. Now, when $i = \text{prog}_{\frac{1}{4}}(x) + 1$, we have that $[\tilde{\nabla}^q F_T(x, z)]_i = \frac{z}{\rho} \cdot (\nabla_i^q F_T(x) + b_i^q(x))$. if $z = 0$
 524 (with probability $1 - \rho$), then we have that $[\tilde{\nabla}^q F_T(x, z)]_i = 0$. So, the first condition follows. Let
 525 $\bar{\nabla}^q F(x)$ be a stochastic but unbiased estimator of $\nabla^q F(x)$. We then have that

$$\begin{aligned} &\mathbb{E}[\|\tilde{\nabla}^q F_T(x, z) - \nabla^q F_T(x)\|^2] \\ &\leq 2\mathbb{E}[\|\tilde{\nabla}^q F_T(x, z) - \bar{\nabla}^q F_T(x, z)\|^2] + 2\mathbb{E}[\|\bar{\nabla}^q F_T(x, z) - \nabla^q F_T(x, z)\|^2] \\ &\leq 2\|b_q(x)\|^2 + \frac{2\ell_{q-1}^2(1-\rho)}{\rho} \\ &\leq 2B_q^2 + \frac{2\ell_{q-1}^2(1-\rho)}{\rho} \end{aligned}$$

526 which finishes the proof. □

527 C.2 Theorem 1 Proof

528 **Remark 1.** Our oracle model is the same as in [3], when setting $M = m = 0$, $\sigma = \sigma_1$, $\zeta = B_1$, and
 529 finding a point x where $\|\nabla F(x)\| = O((\epsilon + B_1^2)^{1/2})$

530 *Proof.* The setting of constants M, m, σ, ζ follows from definition 1, assumption 3, and assumption
 531 4 in [3]. The last point follows from the fact that in Theorem 4 of [3], the goal was to have iterates
 532 $\{x_t\}$ such that

$$\frac{1}{T} \sum_{t=0}^{T-1} \mathbb{E} \|\nabla f(x^{(t)})\|^2 = O(\epsilon + B_1^2)$$

533 which we can equivalently express as

$$\mathbb{E} \|\nabla F(\hat{x})\| = O((\epsilon + B_1^2)^{\frac{1}{2}})$$

534 where $\hat{x}$ is drawn uniformly from $\{x_t\}$. □

535 **Theorem 7.** (Theorem 1 restated). When $p = 1$, there exists $F \in \mathcal{F}_1(\Delta, L_1)$ and $(O_F^1, P_z) \in$
 536 $\mathcal{O}_1(F, \sigma_1, B_1)$ such that for any first-order zero-respecting algorithm (definition 1) where $\epsilon < \frac{1}{4}$ and
 537 $B_1 \leq O(1)$, the minimum number of queries to obtain a $(\epsilon + B_1^2)^{\frac{1}{2}}$ stationary point with constant
 538 probability is bounded below by

$$\Omega\left(\frac{\Delta L_1}{\epsilon + B_1^2} + \frac{\Delta L_1 \sigma_1^2}{\epsilon^2 + B_1^4}\right)$$

539 *Proof.* We let $F_T^* = \alpha F_T(\beta x)$ for some constants α, β which we set in this proof. With probability
 540 at least $\frac{3}{4}$, we have that $\text{prog}(x_{A[O_F^p]}^{(t)}) < T$ for all $t \leq \frac{T-2}{2\rho}$. Since $\text{prog}_1(x) \leq \text{prog}(x)$, we have
 541 that

$$\mathbb{E} \|\nabla F_T^*(x_{A[O_F^p]}^{(t)})\| = \alpha \beta \mathbb{E} \|\nabla F_T(x_{A[O_F^p]}^{(t)})\| \geq \frac{\alpha \beta}{2}$$

542 and that

$$\mathbb{E} \|\tilde{\nabla}^q F_T^*(x, z) - \nabla^q F_T^*(x, z)\|^2 \leq \alpha^2 \beta^{2q} \left(\frac{2\ell_{q-1}^2(1-\rho)}{\rho} + 2B_q^2 \right)$$

543 for all q . Notice that by construction of F_T^* , we have that

- 544 • $F_T^*(0) - \inf_x F_T^*(x) = \alpha(F_T(0) - \inf_x F_T(\alpha x)) \leq \alpha \Delta_0 T$
- 545 • $\|\nabla^2 F_T^*(x)\| = \alpha \beta^{q+1} \|\nabla^2 F_T(\beta x)\| \leq \alpha \beta^2 \ell_1$
- 546 • $\|\nabla F_T^*(x)\| \geq \alpha \beta \|\nabla F_T(x)\| \geq \frac{\alpha \beta}{2}$

547 We also have that

$$\mathbb{E} \|\tilde{\nabla} F_T^*(x, z) - \nabla F_T^*(x, z)\|^2 \leq \alpha^2 \beta^2 \left(\frac{2\ell_0^2(1-\rho)}{\rho} + 2B_1^2 \right)$$

548 We set constants such that

- 549 • $\alpha \Delta_0 T \leq \Delta$
- 550 • $\alpha \beta^2 \ell_1 \leq L_1$
- 551 • $\frac{\alpha \beta}{2} \geq (\epsilon + B_1^2)^{\frac{1}{2}}$
- 552 • $\alpha^2 \beta^2 \left(\frac{2\ell_0^2(1-\rho)}{\rho} + 2B_1^2 \right) \leq 2\sigma_1^2 + 2B_1^2 \implies \alpha^2 \beta^2 \left(\frac{\ell_0^2(1-\rho)}{\rho} + B_1^2 \right) \leq \sigma_1^2 + B_1^2$

553 First, let $\alpha = 2(\epsilon + B_1^2)^{\frac{1}{2}}/\beta$. We then set

$$\rho = \min\left\{\frac{\alpha^2\beta^2\ell_0^2}{\sigma_1^2 + B_1^2 - \alpha^2\beta^2B_1^2}, 1\right\} = \min\left\{\frac{4(\epsilon + B_1^2)\ell_0^2}{\sigma_1^2 + B_1^2 - 4(\epsilon + B_1^2)B_1^2}, 1\right\}$$

554 With this choice of ρ , it's easy to check that

$$\alpha^2\beta^2\left(\frac{\ell_0^2(1-\rho)}{\rho} + B_1^2\right) \leq \sigma_1^2 + B_1^2$$

555 To satisfy the Lipschitz condition, we set

$$\beta = \frac{L_1}{2(\epsilon + B_1^2)^{\frac{1}{2}}\ell_1}$$

556 and we set

$$T = \lfloor \frac{\Delta}{\alpha\Delta_0} \rfloor = \lfloor \frac{\Delta\beta}{2\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}} \rfloor$$

557 By Lemma 9, we have that

$$\begin{aligned} \frac{T-2}{2\rho} &= \frac{1}{2\rho}(\lfloor \frac{\Delta\beta}{2\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}} - 2 \rfloor) \\ &\geq \frac{1}{2\rho} \cdot \frac{\Delta\beta}{4\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}} \\ &\geq \Omega\left(\frac{\sigma_1^2 + B_1^2 - 4(\epsilon + B_1^2)B_1^2}{8(\epsilon + B_1^2)\ell_0^2} \cdot \frac{\Delta}{4\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}} \cdot \frac{L_1}{2(\epsilon + B_1^2)^{\frac{1}{2}}\ell_1}\right) \\ &= \Omega\left(\frac{\Delta L_1(\sigma_1^2 + (1-4\epsilon)B_1^2 - 4B_1^4)}{64\Delta_0\ell_0^2\ell_1(\epsilon + B_1^2)^2}\right) \end{aligned}$$

558 Now, since $\epsilon < \frac{1}{4}$, we can continue to lower bound this expression as follows:

$$\begin{aligned} &\Omega\left(\frac{\Delta L_1(\sigma_1^2 - 4B_1^4)}{64\Delta_0\ell_0^2\ell_1(\epsilon + B_1^2)^2}\right) \\ &\geq \Omega\left(\frac{\Delta L_1\sigma_1^2}{64\Delta_0\ell_0^2\ell_1(\epsilon + B_1^2)^2} - \frac{4\Delta L_1}{64\Delta_0\ell_0^2\ell_1}\right) \\ &\geq \Omega\left(\frac{\Delta L_1\sigma_1^2}{\Delta_0\ell_0^2\ell_1(\epsilon + B_1^2)^2}\right) \\ &= \Omega\left(\frac{\Delta L_1\sigma_1^2}{\Delta_0\ell_0^2\ell_1(\epsilon^2 + B_1^4)}\right) \end{aligned}$$

559 Now, considering the case where the derivative oracles are biased but not stochastic (i.e $\sigma_1 = 0$), we
560 have the following conditions:

- 561 • $\alpha\Delta_0T \leq \Delta$
- 562 • $\alpha\beta^2\ell_1 \leq L_1$
- 563 • $\frac{\alpha\beta}{2} \geq (\epsilon + B_1^2)^{\frac{1}{2}}$
- 564 • $\alpha^2\beta^2\left(\frac{2\ell_0^2(1-\rho)}{\rho} + 2B_1^2\right) \leq 2B_1^2 \implies \alpha^2\beta^2\left(\frac{\ell_0^2(1-\rho)}{\rho} + B_1^2\right) \leq B_1^2$

565 Again, we let $\alpha = 2(\epsilon + B_1^2)^{\frac{1}{2}}/\beta$. We then set

$$\rho = \min\left\{\frac{\alpha^2\beta^2\ell_0^2}{B_1^2(1 - \alpha^2\beta^2)}, 1\right\} = \min\left\{\frac{4(\epsilon + B_1^2)\ell_0^2}{B_1^2 - 4B_1^2(\epsilon + B_1^2)}, 1\right\}$$

566 We again set

$$\beta = \frac{L_1}{2(\epsilon + B_1^2)^{\frac{1}{2}} \ell_1}$$

567 and

$$T = \lfloor \frac{\Delta}{\alpha \Delta_0} \rfloor = \lfloor \frac{\Delta \beta}{2(\epsilon + B_1^2)^{\frac{1}{2}} \Delta_0} \rfloor$$

568 By Lemma 9, we have that

$$\begin{aligned} \frac{T-2}{2\rho} &= \frac{1}{2\rho} (\lfloor \frac{\Delta \beta}{2(\epsilon + B_1^2)^{\frac{1}{2}} \Delta_0} \rfloor - 2) \\ &\geq \frac{1}{2\rho} \cdot \frac{\Delta \beta}{4\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}} \\ &\geq \Omega\left(\frac{\Delta \beta}{\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}}\right) \end{aligned}$$

569 since for all $B_1 \leq O(1)$, $\rho = \Theta(1)$. When we further lower bound this expression, we have that

$$\Omega\left(\frac{\Delta \beta}{\Delta_0(\epsilon + B_1^2)^{\frac{1}{2}}}\right) \geq \Omega\left(\frac{\Delta L_1}{\Delta_0(\epsilon + B_1^2) \ell_1}\right) = \Omega\left(\frac{\Delta L_1}{\epsilon + B_1^2}\right)$$

570 Putting the two lower bound expressions together (as in [20]) yields the matching lower bound:

$$\Omega\left(\frac{\Delta L_1}{\epsilon + B_1^2} + \frac{\Delta L_1 \sigma_1^2}{\epsilon^2 + B_1^4}\right)$$

571

□

572 C.3 Theorem 2 Proof

573 **Theorem 8.** (Theorem 2 restated). For all $p \geq 2$, $\Delta, L_{1:p}, \sigma_{1:p} > 0$, $\epsilon < \sqrt{\sigma_1}$, and $\max_i B_i \leq \frac{\sqrt{3}}{2} \sigma_1$,
 574 there exists $F \in \mathcal{F}_p(\Delta, L_{1:p})$ and $(O_F^p, P_z) \in \mathcal{O}_p(F, \sigma_{1:p}, B_{1:p})$ such that for any p^{th} order zero-
 575 respecting algorithm, the number of queries to obtain a point an $\epsilon + \max_i B_i$ stationary point with
 576 constant probability is bounded below by

$$\begin{aligned} &\Omega(1) \cdot \frac{(\sigma_1^2 - 4(\epsilon + B)^2 B_1^2) \Delta}{32(\epsilon + B)^3 \ell_0^2 \Delta_0} \cdot \\ &\min_{q' \in \{1, \dots, p\}} \min_{q \in \{2, \dots, p\}} \left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B) \ell_{q'}} \right)^{\frac{1}{q'}} \right\} \end{aligned}$$

577 *Proof.* We perform a similar argument to that for the proof of Theorem 1, except now accounting for
 578 the higher order Lipschitz constraints. We have that

$$\mathbb{E} \|\nabla F_T^*(x_{A[O_F^p]}^{(t)})\| = \alpha \beta \|\nabla F_T(x_{A[O_F^p]}^{(t)})\| \geq \frac{\alpha \beta}{2}$$

579 and that

$$\mathbb{E} \|\tilde{\nabla}^q F_T^*(x, z) - \nabla^q F_T^*(x, z)\|^2 \leq \alpha^2 \beta^{2q} \left(\frac{2\ell_{q-1}^2 (1 - \rho)}{\rho} + 2B_q^2 \right)$$

580 We now set constants such that

- 581 • $\alpha \Delta_0 T \leq \Delta$
- 582 • $\alpha \beta^{q+1} \ell_q \leq L_q$
- 583 • $\frac{\alpha \beta}{2} \geq \epsilon + \max_j B_j$

$$\bullet \alpha^2 \beta^{2q} \left(\frac{2\ell_{q-1}^2(1-\rho)}{\rho} + 2B_q^2 \right) \leq 2\sigma_q^2 + 2B_q^2 \implies \alpha^2 \beta^{2q} \left(\frac{\ell_{q-1}^2(1-\rho)}{\rho} + B_q^2 \right) \leq \sigma_q^2 + B_q^2$$

First, let

$$\alpha = \frac{2(\epsilon + \max_j B_j)}{\beta}$$

Now, we set

$$\rho = \min\left\{ \frac{\alpha^2 \beta^2 \ell_0^2}{\sigma_1^2 + B_1^2 - \alpha^2 \beta^2 B_1^2}, 1 \right\}$$

This implies that

$$\begin{aligned} & \alpha^2 \beta^{2q} \left(\frac{\ell_{q-1}^2(1-\rho)}{\rho} + B_q^2 \right) \\ & \leq \alpha^2 \beta^{2q} \left(\frac{\ell_{q-1}^2}{\rho} + B_q^2 \right) \\ & \leq \alpha^2 \beta^{2q} \left(\frac{\ell_{q-1}^2(\sigma_1^2 + B_1^2 - \alpha^2 \beta^2 B_1^2)}{\alpha^2 \beta^2 \ell_0^2} + B_q^2 \right) \\ & \leq \frac{\beta^{2(q-1)} \ell_{q-1}^2 (\sigma_1^2 + B_1^2 - \alpha^2 \beta^2 B_1^2)}{\ell_0^2} + \alpha^2 \beta^{2q} B_q^2 \\ & = \frac{\beta^{2(q-1)} \ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)}{\ell_0^2} + \alpha^2 \beta^{2q} B_q^2 \end{aligned}$$

where $B = \max_j B_j$. Letting

$$\frac{\beta^{2(q-1)} \ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)}{\ell_0^2} + \alpha^2 \beta^{2q} B_q^2 \leq \sigma_q^2 + B_q^2$$

and solving for β such that

$$\frac{\beta^{2(q-1)} \ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)}{\ell_0^2} \leq \frac{\sigma_q^2 + B_q^2}{2}$$

and

$$\alpha^2 \beta^{2q} B_q^2 \leq \frac{\sigma_q^2 + B_q^2}{2}$$

and the L_q -condition holds, yields

$$\beta = \min_{q' \in \{1, \dots, p\}, q \in \{2, \dots, p\}} \min\left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B)\ell_{q'}} \right)^{\frac{1}{q'}} \right\}$$

Setting

$$T = \lfloor \frac{\Delta}{\alpha \Delta_0} \rfloor = \lfloor \frac{\Delta \beta}{2\Delta_0(\epsilon + B)} \rfloor$$

We now have that (assuming $T \geq 5$)

$$\begin{aligned} \frac{T-2}{2\rho} &= \frac{1}{2\rho} (\lfloor \frac{\Delta \beta}{2\Delta_0(\epsilon + B)} \rfloor - 2) \\ &\geq \frac{1}{2\rho} \cdot \frac{\Delta \beta}{4\Delta_0(\epsilon + B)} \\ &\geq \frac{\sigma_1^2 + B_1^2 - \alpha^2 \beta^2 B_1^2}{2\alpha^2 \beta^2 \ell_0^2} \cdot \frac{\Delta}{4\Delta_0(\epsilon + B)} \\ &\quad \min_{q' \in \{1, \dots, p\}, q \in \{2, \dots, p\}} \min\left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B)\ell_{q'}} \right)^{\frac{1}{q'}} \right\} \\ &\geq \frac{(\sigma_1^2 - 4(\epsilon + B)^2 B_1^2) \Delta}{32(\epsilon + B)^3 \ell_0^2 \Delta_0} \\ &\quad \min_{q' \in \{1, \dots, p\}, q \in \{2, \dots, p\}} \min\left\{ \left(\frac{\ell_0^2 (\sigma_q^2 + B_q^2)}{2\ell_{q-1}^2 (\sigma_1^2 + B_1^2 - 4(\epsilon + B)^2 B_1^2)} \right)^{\frac{1}{2(q-1)}}, \left(\frac{\sigma_q^2 + B_q^2}{8(\epsilon + B)^2 B_q^2} \right)^{\frac{1}{2(q-1)}}, \left(\frac{L_{q'}}{2(\epsilon + B)\ell_{q'}} \right)^{\frac{1}{q'}} \right\} \end{aligned}$$

594 which finishes the proof.

595

□

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